

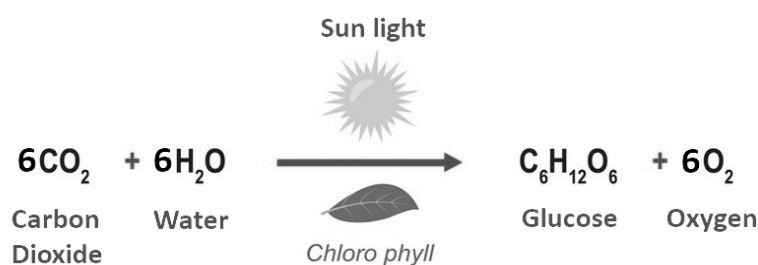
BIOLOGY CLASS NOTES PART 2

6- PLANT

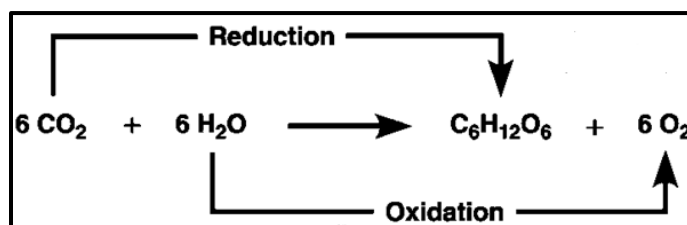
Plants are vital **multicellular organisms** that form the basis of most ecosystems. As autotrophs, they produce food by photosynthesis, releasing oxygen and supporting life. They also undergo transpiration, the loss of water through the leaves, which aids cooling and nutrient transport. Plants require macronutrients (N, P, K) and micronutrients (Fe, Zn). Xylem transports water and minerals, while phloem distributes food. Growth and development are regulated by plant hormones, and reproduction occurs **sexually or asexually**.

1. Photosynthesis

- **Jan Ingenhousz**, a Dutch-born British physician and scientist, discovered that green plants in sunlight absorb **carbon dioxide** and release **oxygen** during **photosynthesis**.
- **Photosynthesis** is the process by which phototrophs convert **light energy into chemical energy**, stored as **glucose**, using **carbon dioxide and water in the presence of sunlight**.



- The process occurs mainly in **chloroplasts** with the help of pigments such as **chlorophyll, carotene, and xanthophyll**. Oxygen released during photosynthesis is used by animals and humans for **respiration**.



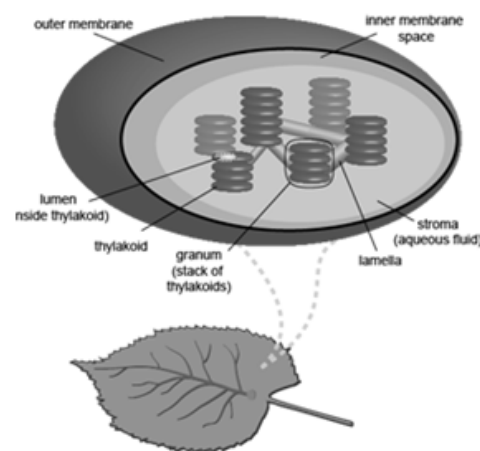
1.1. How Does Photosynthesis Take Place?

- Photosynthesis occurs in two stages:
 1. **Light Reaction** (in the presence of sunlight)
 2. **Dark Reaction** (light-independent stage)

Photosynthesis In Presence Of Sunlight (Light Reaction)

- Photosynthesis takes place primarily in the **chloroplasts of green leaves**. **Leaves are the food factories of plants**.
- The leaves have a green pigment called **chlorophyll**. It helps leaves to capture the energy of the sunlight.

Photosynthesis in the presence of sunlight takes place in the following steps:



- Water and minerals are transported from the soil to the leaves through **xylem vessels**, which form a continuous pathway through the roots, stem, branches, and leaves.
- **Chloroplasts** are small organelles in plant cells where photosynthesis occurs and sunlight is converted into chemical energy.
- Chloroplasts usually align along the walls of **mesophyll cells** to receive optimum light.
- A chloroplast contains a membranous system of **grana** and a matrix called **stroma**.
- The sac-like structures called **thylakoids** contain **chlorophyll**, the light-absorbing pigment responsible for the green colour of plants.
- During photosynthesis, chlorophyll absorbs mainly **blue and red light** and reflects **green light**.
- The chloroplast shows division of labour: the membrane system traps light energy and synthesizes **ATP and NADPH**.
- The main function of the light reaction is to produce **ATP and NADPH**, which are required for the **subsequent dark reaction**.

Photosynthesis in Artificial light

- Photosynthesis can even occur in artificial light if the plant is exposed to the **correct wavelength of light**.
- The maximum photosynthesis occurs between **600-700 nm in the red region**, which is followed by **400-500 nm in the blue region**, and the **least between 500-600 nm in the green region**.

BCW Fact: **Photolysis** is the process by which light energy breaks down a molecule into smaller parts. In photosynthesis, photolysis refers to the **splitting of water molecules into hydrogen and oxygen**.

Dark Reaction

- The **dark reaction** or **light-independent stage**, also called the **Calvin Cycle**, occurs in the **stroma** of chloroplasts.
- In the stroma, enzymatic reactions synthesize **sugar**, which is later converted into **starch**.
- **Carbon dioxide** enters the leaves through tiny pores called **stomata**, which are surrounded by **guard cells**.
- The dark reaction uses **ATP and NADPH** produced during the light reaction to convert **six molecules of carbon dioxide into one glucose molecule**.
- This reaction is also known as the **C3 cycle** or **reductive pentose phosphate cycle**.
- Key enzymes involved are **Rubisco**, **phosphoglycerate kinase (PGK)**, **glyceraldehyde-3-phosphate dehydrogenase (GAPDH)**, and **ribulose-5-phosphate kinase (Ru5PK)**.
- **Rubisco** catalyses the first step of carbon fixation, while **PGK** and **GAPDH** convert 3-phosphoglycerate into glyceraldehyde-3-phosphate. **Ru5PK** converts ribulose-5-phosphate into ribulose-1,5-bisphosphate.

NOTE:

- The **dark reactions of photosynthesis (Calvin Cycle)** in **cyanobacteria** occur in the **cytosol**.
- Presence of both **DNA & RNA** in chloroplasts enables them to self-replicate.

BCW Fact: **C3** and **C4 plants** are classified based on the pathways they use during the dark reaction of photosynthesis. About 95% of plants, including most trees and shrubs, are C3 plants and lack **Kranz anatomy** in their leaves. In contrast, C4 plants use the **Hatch-Slack (C4) pathway** during the dark reaction and constitute about 5% of plant species on Earth.

2. Transpiration

- **Transpiration** is the **evaporative loss of water from plants**, occurring mainly through the **aerial parts** such as **leaves, stems, and flowers**. Besides water loss, **exchange of oxygen (O₂) and carbon dioxide (CO₂)** also occurs through **stomata**. Normally, stomata remain open during the day and close at night.

Stomatal Transpiration

- **Stomatal transpiration** refers to the evaporation of water through **stomata**.
- It accounts for **most of the water loss in plants**.
- Water present near the leaf surface converts into **vapour and escapes when stomata are open**.

Factors Affecting Transpiration

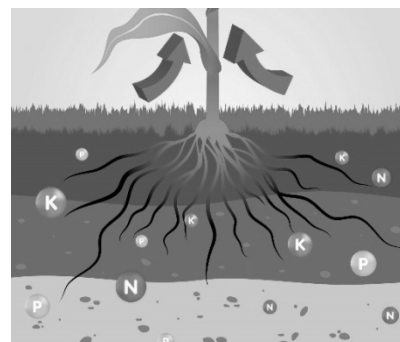
- **External Factors**
 - Temperature
 - Light intensity
 - Humidity
 - Wind speed
- **Plant Factors**
 - Number and distribution of stomata
 - Percentage of open stomata
 - Water status of the plant
 - Canopy structure

Significance of Transpiration

1. Helps in the **transport of water and minerals** to different plant parts.
2. Maintains **water balance** in the plant body.
3. Maintains **osmosis** and keeps cells **turgid (rigid)**.
4. Generates **transpiration pull**, aiding the **upward movement of water**.
5. In some plants, **hydrophilic salts** accumulate on leaf surfaces and help retain moisture.

3. Macronutrients And Micronutrients Of Plant

- Plants absorb **inorganic elements (ions) and salts** mainly from **soil and water**. Nutrients move across cell membranes by:
 - **Diffusion**
 - **Facilitated transport**
 - **Active transport**
- **Transport system**
 - **Xylem**: Transports **water and minerals**
 - **Phloem**: Transports **organic food materials**
- Essential elements are classified into:
 1. **Macronutrients** → Required in **large amounts**
 2. **Micronutrients** → Required in **small amounts**



3.1. Macronutrients

They are generally present in plant tissues in large amounts. The macronutrients include **carbon, hydrogen, oxygen, nitrogen, phosphorus, sulphur, potassium, calcium and magnesium**.

1. Nitrogen:

- Nitrogen is required by all parts of a plant, particularly the meristematic tissues and the metabolically active cells. Nitrogen is one of the major **constituents of proteins, nucleic acids, vitamins, and hormones**. **Insectivorous plants grow in nitrogen deficient soil**. They trap insects to compensate for nitrogen deficiency.

2. Phosphorus:

- Phosphorus is absorbed by the plants from soil in the form of phosphate ions.
- It is a **constituent of cell membranes, certain proteins, all nucleic acids and nucleotides**, and is required for all phosphorylation reactions.
- **Phosphorus deficiency** tends to inhibit or **prevent shoot growth**. Leaves turn dark, dull, **reddish, or reddish-violet** colour due to increased anthocyanin synthesis.



3. Potassium:

- Potassium helps to maintain an anion-cation balance in cells and is involved in protein synthesis, **opening and closing of stomata, activation of enzymes**, and maintenance of the **turgidity of cells**.

4. Calcium:

- Plants absorb calcium from the soil in the form of calcium ions (Ca^{2+}). Calcium is **required by meristematic and differentiating tissues**. During cell division, it is used in the **synthesis of the cell wall**.

5. Magnesium:

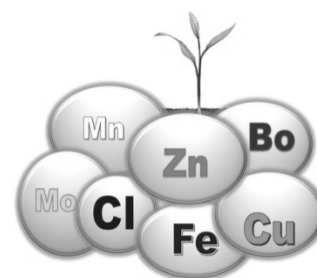
- Magnesium (Mg) is a constituent of chlorophyll molecules and is absorbed by plants as Mg^{2+} ions. It activates several enzymes **involved in respiration and photosynthesis**, participates in **DNA and RNA synthesis**, and helps maintain the structural integrity of ribosomes.

3.2. Micronutrients

Micronutrients or trace elements are needed in **very small quantities**. These include **iron, manganese, copper, molybdenum, zinc, boron, chlorine and nickel**.

Since each element has one or **more specific structural or functional roles** in plants, in the absence of any particular element, plants show certain morphological changes. These morphological changes are indicative of certain element deficiencies.

NOTE: Sodium (Na) is a notable absentee in Micronutrients group.



4. Xylem And Phloem

Xylem and phloem are the two vascular tissues responsible for transport in plants. Together, they form the **vascular bundle**. Transport in **xylem is unidirectional** (root to shoot), whereas transport in **phloem is bidirectional**.

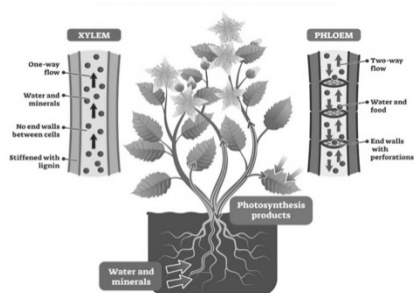
4.1. Xylem

- Xylem transports **water and dissolved minerals** from roots to stems and leaves.
- It is composed of **tracheids, vessels, xylem parenchyma, and xylem fibres**. All xylem cells are dead except **xylem parenchyma**.
- The transport of xylem sap (water and minerals) is **passive**, driven mainly by **transpiration pull** and aided by **capillary action**.
- **Capillary action** and **transpiration pull** facilitate the upward movement of xylem sap against gravity. Capillary action is the upward movement of liquid due to **surface tension**.
- In most vascular bundles (e.g., dicot stems), **xylem is located on the inner side**, while **phloem lies on the outer side**.

4.2. Phloem (Bast Tissue)

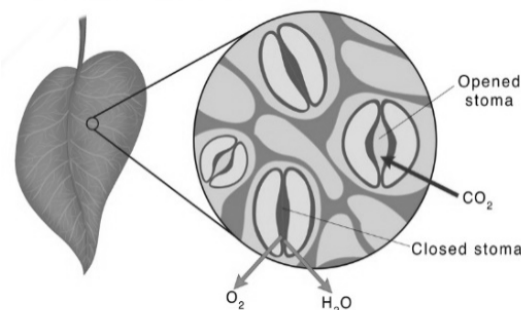
- Phloem aids in the movement of the **photosynthesized food of leaves** (generally in the form of sucrose) to other parts of the plant via the system of translocation. This type of **movement occurs from the leaves to the other parts of the plant**.

- Transport occurs due to an **osmotic pressure gradient** and can be **bidirectional**.
- Phloem consists of **sieve tubes, sieve cells, companion cells, phloem parenchyma, and phloem fibres (sclerenchyma)**.
- In most vascular bundles, phloem lies **outside the xylem**.
- Damage to the phloem (e.g., removal of bark) disrupts food transport and may lead to the death of the plant.



5. Stomata

- **Stomata** are minute pores present in the epidermis of leaves that facilitate **gaseous exchange** and regulate **transpiration**.
- During photosynthesis, **CO₂ enters** the leaf through stomata, while **O₂ and water vapour exit** through them.
- Each stoma is surrounded by a pair of **guard cells**, which control the opening and closing of the stomatal pore.
- **Absorption of water** causes guard cells to swell and open the pore, while **loss of water** causes them to shrink and close it.
- Guard cells contain **chloroplasts** and chlorophyll. They are **dumbbell-shaped in monocots** and **bean-shaped in dicots**.
- **Subsidiary cells** are specialized epidermal cells surrounding the guard cells. They provide support and protection during the opening and closing of stomata.



BCW Fact: Food of the plant gets stored in the form of starch/glucose in roots (called Root Tuber) or stem (called Stem Tuber). Some examples: **Root Tubers:** Carrot, Beetroot, Radish, etc. and **Stem Tubers:** Potato, Yam.

Crescograph

- It is a device used to measure the growth of plants. It was invented by the Indian botanist **Sir Jagadish Chandra Bose**. Two important things used in the apparatus are a smoked glass plate and a number of clockwise gears.

6. Plant Hormones

- Plant hormones, also known as **phytohormones**, are chemical compounds present in very low concentrations in plants.
- They regulate plant development, growth, longevity and reproductive processes. These hormones are produced in almost all parts of the plant and are transmitted to various parts of the plant.

- Plant hormones control all the growth and development activities like cell division, enlargement, flowering, seed formation, dormancy and abscission.

Based on their action, plant hormones are categorized into two categories:

- Plant Growth Promoters
- Plant Growth Inhibitors

7. Plant Growth Promoters

7.1. Auxins

- Auxins are a group of naturally occurring and artificially synthesised plant hormones. They play an important role in the regulation of plant growth. Auxins were first isolated from human urine.
- The term 'auxin' is applied to the indole-3-acetic acid (IAA), and to other natural and synthetic compounds having certain growth regulating properties.
- **Naturally occurring auxins:** IAA and indole butyric acid (IBA) have been isolated from plants.
- **Synthetic auxins:** NAA (naphthalene acetic acid) and 2, 4-D (2, 4-dichlorophenoxyacetic acid) are synthetic auxins. All these auxins have been used extensively in agricultural and horticultural practices.

Functions Of Auxins

1. **Initiate Rooting in Stem Cuttings:** They help initiate rooting in stem cuttings, an application widely used for plant propagation.
2. **Flowering: Auxins promote flowering,** e.g., in pineapples.
3. **Prevention:** They help to **prevent fruit and leaf drop at early stages** but promote the abscission of older mature leaves and fruits.
4. **Apical Dominance:** When the growth of the apical meristem inhibits the **growth of axillary buds**, the phenomenon is known as **apical dominance**. When the shoot tip is removed, it induces the growth of lateral buds. This is used to promote branching, e.g., in hedge-making and tea plantations.
5. **Parthenocarp:** When auxin is applied to some flowers, it induces parthenocarp, i.e., the ovary enlarges and develops into seedless fruit (unfertilized). Seedless tomatoes are widely produced by this method.
6. **Flowering:** It delays the **senescence of flowers**. A high concentration of auxin promotes femaleness in some of the plants. It promotes flowering in litchi and pineapples.
7. **Herbicides:** They are widely used as herbicides. **Synthetic auxins, e.g., 2, 4-D,** are widely used to kill **dicotyledonous weeds** and do not affect mature monocotyledonous plants. It is used to prepare weed-free lawns by gardeners.
8. **Cell Division:** Auxin also **controls xylem** differentiation and helps in cell division.

7.2. Gibberellins

Gibberellins are another kind of promotory Plant Growth Regulator (PGR).

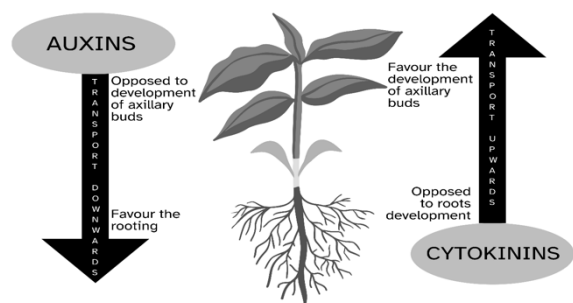
More than 100 gibberellins are reported from widely different organisms such as fungi and higher plants.

- All gibberellins (**GAs**) are acidic in nature, which produces a wide range of **physiological responses in plants**. Their ability to cause an increase in the length of the axis is used to increase the length of grapes stalks. Gibberellins cause fruits like apples to elongate and improve their shape.
- They also **delay senescence**. Thus, the fruits can be left on the tree longer **to extend the market period**. GA₃ is used to **speed up the malting process in the brewing industry**.
- Gibberellins also **promote bolting** (internode elongation just prior to flowering) in beet, cabbage, and many plants with rosette habit.

7.3. Cytokinins

Cytokinins have specific effects on cytokinesis (division of cytoplasm during cell division) and were **discovered as kinetin** (a modified form of adenine, a purine) from autoclaved herring sperm DNA. Kinetin does not occur naturally in plants.

- Since the discovery of zeatin, several naturally occurring cytokinins and some synthetic compounds with **cell-division promoting activity** have been identified.
- Natural cytokinins are **synthesised in regions where rapid cell division** occurs, for example, root apices, developing shoot buds, young fruits, and similar tissues.
- They help **produce new leaves, chloroplasts in leaves, lateral shoot growth**, and adventitious shoot formation. Cytokinins help overcome apical dominance.
- **Cytokinins promote cell division and cell expansion during the growth stage of plants.**
- They promote nutrient mobilisation, which **helps in the delay of leaf senescence.**



8. Plant Growth Inhibitors

8.1. Abscisic Acid

Abscisic acid (ABA) was discovered for **its role in regulating abscission and dormancy**. It acts as a general **plant growth inhibitor and an inhibitor of plant metabolism**.

- It inhibits seed germination.
- It **stimulates the closure of stomata in the**

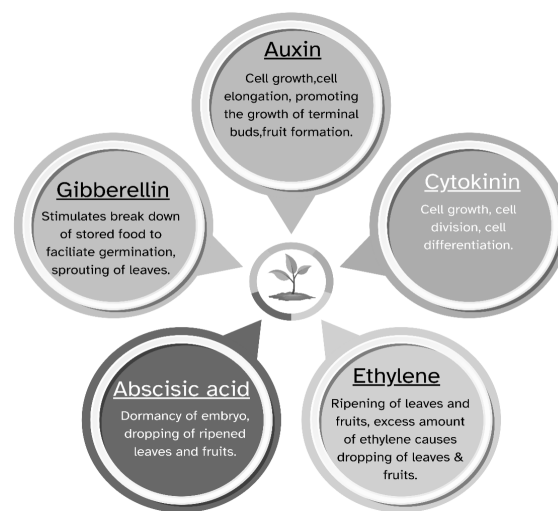
epidermis and increases the tolerance of plants to various kinds of stresses. Therefore, it is also called the stress hormone.

- It plays an important role in **seed development, maturation, and dormancy.**

8.2. Ethylene

Ethylene is a simple gaseous Plant Growth Regulator (PGR).

- Ethylene is highly **effective in fruit ripening**.
- It is **synthesised in large amounts by tissues** undergoing senescence and ripening fruits. It enhances the respiration rate during ripening of the fruits. This rise in the rate of respiration is called **respiratory climacteric**.
- Influences of ethylene on plants include **horizontal growth of seedlings**, swelling of the axis, and apical hook formation in dicot seedlings.
- Ethylene promotes **senescence and abscission** of plant organs, especially of leaves and flowers.
- Ethylene breaks seed and bud dormancy, initiates germination in peanut seeds, and sprouting of potato tubers.



9. Plant Reproduction

- Reproduction is the biological process by which organisms produce offspring. It ensures the continuity of life and involves the replication of **DNA**, which carries hereditary information.
- DNA directs **protein synthesis**, enabling the formation and functioning of an organism's body.

- **Types of Reproduction:**
 1. Asexual Reproduction
 2. Sexual Reproduction

9.1 Asexual Reproduction

Asexual reproduction occurs without the involvement of another organism and produces genetically identical offspring.

Types of Asexual Reproduction:

- **Fission:** A unicellular organism divides into two equal halves (e.g., bacteria, protozoa). New individuals are **created by cell division or fission**.
- **Budding:** A small outgrowth (bud) develops on the parent and detaches when mature (e.g., Hydra).
- **Fragmentation:** The parent body breaks into fragments, each developing into a new individual (e.g., Spirogyra).
- **Regeneration:** New organisms develop from body parts through specialized cells (e.g., Hydra, Planaria).
- **Vegetative Propagation:** New plants arise from vegetative parts such as roots, stems, or leaves. It may occur naturally or be induced artificially in horticulture. Offspring are genetically identical to the parent, maintaining 100% genetic purity.

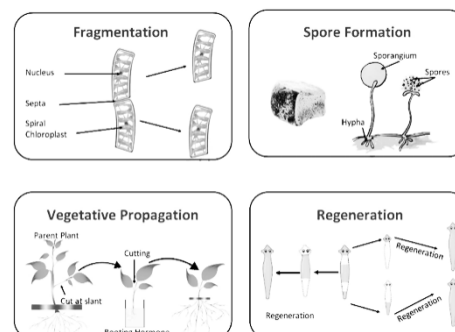


	Germination	Growth to Maturity	Flowering	Fruit Development	Abscission	Seed Dormancy
Gibberellin	✓	✓	✓	✓	✗	✗
Auxin	✗	✓	✓	✓	✗	✗
Cytokinins	✗	✓	✓	✓	✗	✗
Ethylene	✗	✗	✓	✓	✓	✗
Abscissic Acid	✗	✗	✗	✗	✓	✓

9.2 Sexual Reproduction

Sexual reproduction involves the fusion of male and female gametes, each containing half the normal chromosome number.

- Offspring inherit genetic material from both parents.
- DNA copying during reproduction introduces variations.
- Variations form the basis of evolution and help species adapt to changing environments.
- 100% Genetic purity is not maintained, as genes are contributed by both parents.

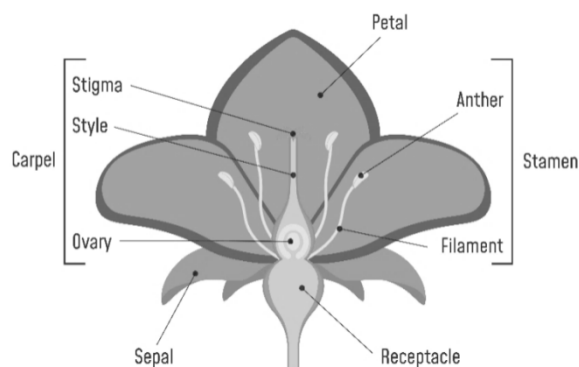


9.3 Reproduction in Plants

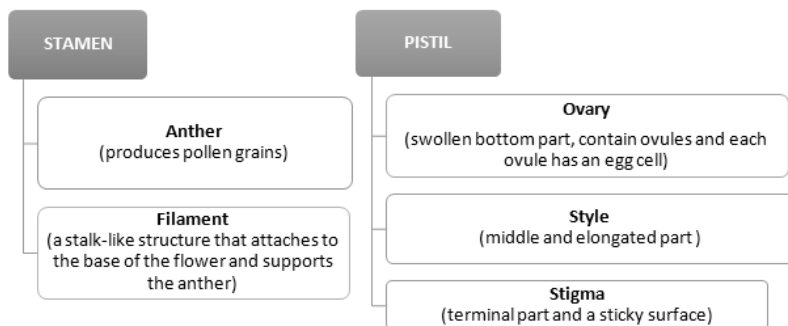
- Flowering plants (angiosperms) reproduce sexually through flowers, which contain reproductive organs.
- **Stamen:** Male reproductive organ; anthers produce pollen grains.
- **Pistil (Carpel):** Female reproductive organ consisting of stigma, style, and ovary.
- **Ovules** within the ovary produce female gametes (eggs).
- Fusion of male and female gametes forms a **zygote**.

Types of Flowers:

- **Unisexual:** Possess either stamens or pistils (e.g., Papaya).
- **Bisexual:** Possess both stamens and pistils (e.g., Rose).



9.3.1 Pollination



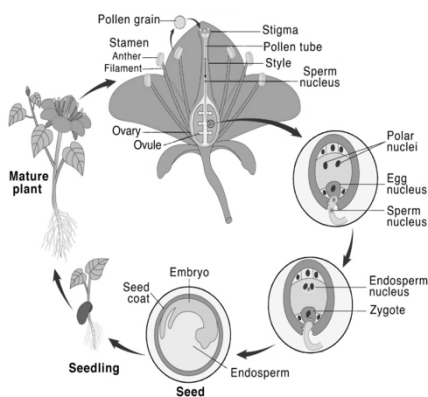
flower of the same plant, or that of a different plant of the same kind. The process is commonly done by insects and wind.

9.3.2. Fertilisation

The fusion of the **germ cells**, or **fertilisation**, forms the **zygote**, which can grow into a new plant.

Fertilization in a flower involves the fusion of male cells carried by **pollen grains with egg cells in an ovule to form a zygote**.

- After fertilisation, the **zygote divides several times** to form an embryo within the ovule. The Embryo eventually **develops into seeds of fruit**.
- The ovary grows rapidly and ripens **to form a fruit**. The seed contains **the future plant or embryo**, which develops into a seedling under appropriate conditions. This is **known as Germination**.



BCW Facts:

- **Azolla** is a plant that is used as a biofertilizer these days as it has **nitrogen-fixing bacteria (cyanobacteria Anabaena Azollae)** that fix free nitrogen, leading to increased soil productivity.
- **Halophytes** are specialized plants that have adapted to thrive in environments with **high salinity (i.e., salt-loving plant)**. **Xerophytes** are plants that are capable of growing in **extremely dry conditions** throughout the entire year.
- The **epidermis** constitutes the outermost layer of the primary plant body. Typically, the **epidermis is a single layer thick**. The cells of the epidermis are **parenchymatous**, containing a minimal amount of cytoplasm adjacent to the cell wall and a large vacuole.
- Climbers and creepers are plants with soft, flexible stems. **Climbers** grow upward by attaching themselves to other structures or plants for support, while **creepers** spread horizontally along the ground. Examples of climbers include **Money Plant, Jasmine**, and **Bottle Gourd**, whereas **Watermelon** and **Pumpkin** are typical examples of creepers.
- The **petiole**, a part of the leaf, attaches the leaf blade to the stem, often with a leaf base at the point of attachment. It also helps the leaf blade access light and allows it to flutter in the wind.

7 – CENTRAL NERVOUS SYSTEM

The **Central Nervous System (CNS)** consists of the **brain and spinal cord** and serves as the body's control centre. The brain is divided into the **forebrain, midbrain, and hindbrain**, which regulate cognition, balance, and involuntary functions. The **cerebrum** controls intelligence and voluntary actions, while the **cerebellum** and **medulla oblongata** coordinate movement and regulate vital reflexes. The **hypothalamus**, a part of the forebrain, links the nervous and endocrine systems by controlling the **pituitary gland**, which secretes hormones such as **TSH, ACTH, oxytocin, and ADH**. Together, the CNS and endocrine system regulate growth, metabolism, reproduction, and homeostasis, enabling the body to maintain internal balance and respond to environmental changes.

1. Neural System

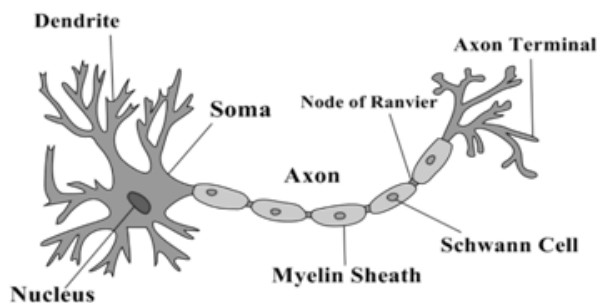
- The neural system of all animals is composed of **highly specialised cells called neurons**, which can detect, receive, and transmit different kinds of stimuli. The human neural system is divided into two parts: the central neural system (CNS) and the peripheral neural system (PNS).
- The CNS includes the **brain and the spinal cord** and is the **site of information processing and control**. The PNS comprises **all the nerves of the body associated with the CNS** (brain and spinal cord).

2. NEURON

- Neurons** are the structural and functional units of the nervous system.
- They receive sensory information, process it, and transmit electrical impulses to muscles, glands, and other neurons.

Parts of a Neuron

- A neuron consists of **three main parts: cell body (soma), dendrites, and axon**.
- Cell Body (Soma):** Contains the nucleus and genetic material; maintains the neuron's structure and provides energy for cellular activities.
- Dendrites:** Short, branched projections containing **Nissl's granules** that receive signals from other neurons and transmit them towards the cell body.
- Axon:** A long fibre that carries nerve impulses away from the cell body to other neurons, muscles, or glands. It is the output region of the neuron.



3. Central Neural System

3.1. Brain

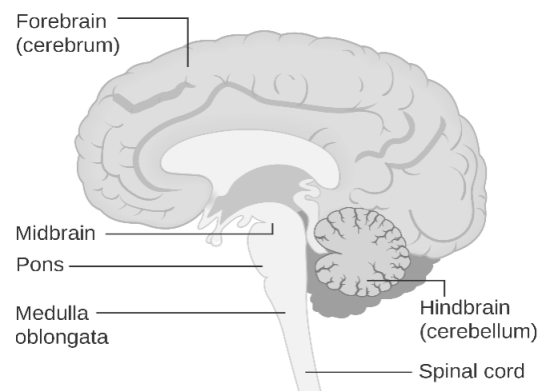
- The **brain** is the central information-processing organ and the **command and control centre** of the body. It weighs approximately **1.4 kg**.
- It controls **voluntary movements, body balance**, and the functioning of vital involuntary organs such as the **heart, lungs, and kidneys**.
- It regulates **body temperature, hunger, thirst, circadian rhythms**, and the activities of several **endocrine glands**.
- It is the centre for **vision, hearing, speech, memory, intelligence, emotions, and thought processes**.

Reflex Actions

- These are sudden and involuntary actions in response to something in the environment. It does not involve the conscious part of the brain.
- A connection, called a **reflex arc**, is formed in the spinal cord, when nerves from all over the body meet in a bundle (**31 pairs of nerves arises from the spinal cord**).
- The reflex arc speeds up the **process of detecting the signal** and responding to it by an **output action quickly**, although the information input also goes to the brain.
- **Transfer of electrical signals by nerve cells** in the human body is enabled by **Sodium and Potassium ions**.

3.2. Skull

- The **brain is protected by the skull**, which consists of **22 bones—8 cranial bones and 14 facial bones**.
- It lies within the **cranium** and is surrounded by **cerebrospinal fluid (CSF)**.
- The brain lacks **pain receptors**, allowing some brain surgeries to be performed while the patient remains conscious.
- The **Cerebrospinal Fluid (CSF)** is a fluid that circulates within the skull and spinal cord, filling up hollow spaces on the surface of the brain.
- The brain can be divided into three major parts:
 - Forebrain
 - Midbrain
 - Hindbrain

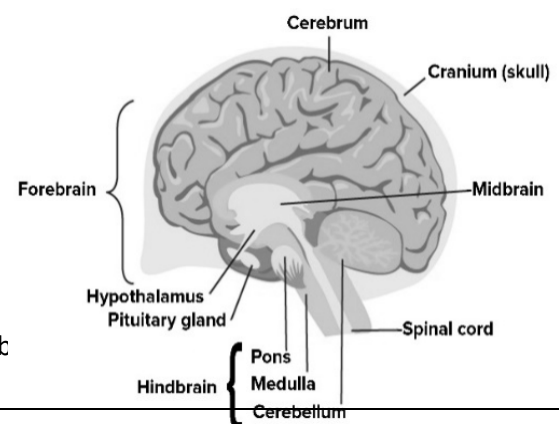


4. Anatomy Of Brain Parts

4.1 Forebrain

- The forebrain consists of the **cerebrum, thalamus, and hypothalamus**. The cerebrum forms the major part of the human brain.
- A deep cleft divides the cerebrum longitudinally into two halves, which are termed the left and right cerebral hemispheres.

(a) Cerebrum



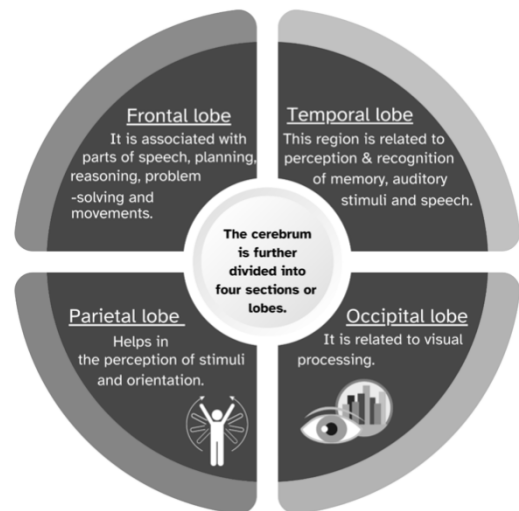
- The cerebrum is the **largest part of the brain (occupies two-thirds of the total brain)**.
- It controls **thinking, intelligence, consciousness, memory, emotions, and sensory perception** (touch, hearing, and vision).
- It consists of **two cerebral hemispheres** connected by a dense band of nerve fibres called the **corpus callosum**.

(b) Thalamus

- The thalamus is a small structure located right above the brain stem, responsible for relaying sensory information from the **sense organs**. It is also responsible for transmitting motor information for movement and coordination.

(c) Hypothalamus

- The hypothalamus lies at the base of the thalamus. It regulates **body temperature, hunger, thirst, sleep, circadian rhythm, emotions, and autonomic functions**, and produces several **hypothalamic hormones**.
- Together with the limbic system—which includes structures like the amygdala and hippocampus—it also controls **sexual behaviour, emotions such as fear and pleasure, and motivation**.
- **The pituitary gland** is the master gland, and the hypothalamus controls and instructs the pituitary gland; hence it is called **super master or master of the master gland**.
- It forms the hypothalamic-pituitary axis, the main link between the nervous and the endocrine systems.



4.2 Midbrain

- It is located **between the thalamus and hypothalamus of the forebrain and pons of the hindbrain**.
- It is involved in the processing of visual and auditory information.
- A canal called the cerebral aqueduct passes through the midbrain. It basically transfers the information from the ear to the brain.

4.3 Hindbrain

- Located in the **lower back region of the brain** and consists of the **cerebellum, medulla oblongata, and pons**.

(a) Cerebellum

- The **second-largest part of the brain**, located posterior to the pons and above the medulla.
- Coordinates **voluntary movements**, maintains **balance, posture, and muscle coordination**.

(b) Medulla Oblongata

- The **lowermost part of the brain**, connecting the **brain and spinal cord**.
- Controls **involuntary functions** such as breathing, heartbeat, blood pressure, digestion, and gastric secretions.

(c) Pons

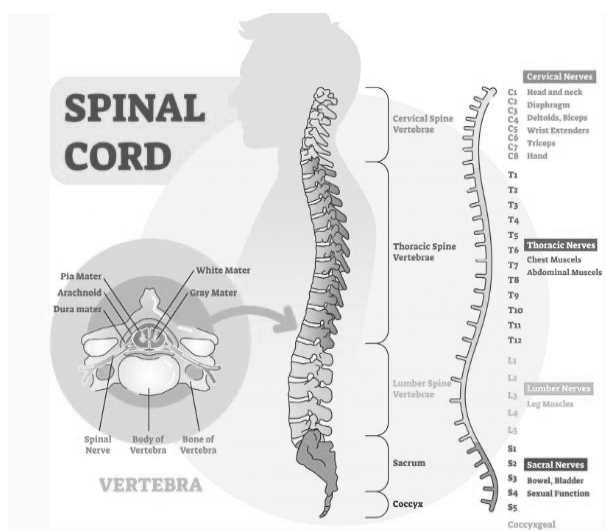
- Part of the **brainstem**, located between the **midbrain and medulla oblongata**.
- Regulates **breathing, sleep-wake cycle, and other unconscious activities**.
- Acts as a **relay centre** for nerve signals between different parts of the brain and the face.

5. Spinal Cord (Governing Centre of Reflex Action)

- The **spinal cord** is a part of the **central nervous system**. It is a long, pipe-like structure arising from the medulla oblongata, part of the brain, consisting of a **collection of nerve fibres, running through the vertebral column of the backbone**.
- In adults, the spinal cord is **usually around 40-45 cm long and 1-1.5 cm wide**. It forms a **vital link between the brain and the body**.
- Divided into five different parts - Cervical cord, Thoracic cord, Lumbar cord, Sacral cord, and Coccygeal cord.
- The spinal cord is divided into **31 segments**, each giving rise to a pair of spinal nerves branching from the spinal cord. There are **8 pairs of cervical nerves, 12 pairs of thoracic nerves, 5 pairs of lumbar nerves, 5 pairs of sacral nerves and 1 pair of coccygeal spinal nerves**.
- It performs the primary processing of information as it carries sensory signals from all parts of the body to the Central Nervous System through afferent fibres.

6. Endocrine System

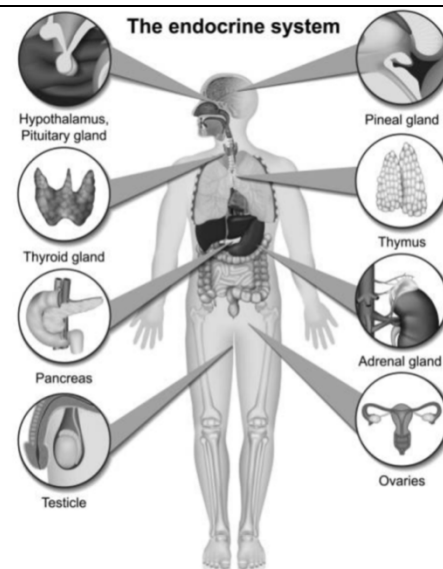
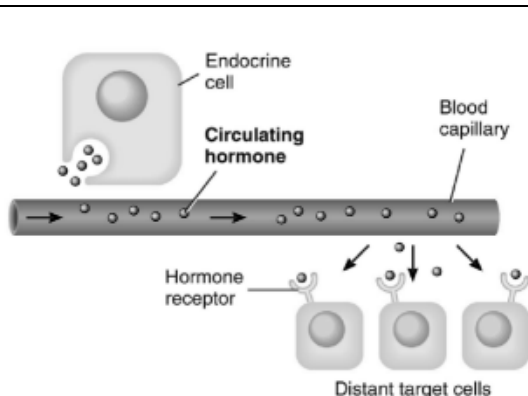
- The **endocrine system** consists of several glands distributed throughout the body.
- Endocrine glands** are **ductless glands** that secrete **hormones** directly into the bloodstream.
- Hormones** are **non-nutrient chemical messengers** produced in trace amounts that regulate various body functions.
- The term "**hormone**" was introduced by Ernest Henry Starling in **1905**.



	Endocrine Glands	Exocrine Glands
Duct	Endocrine glands lack ducts and are hence called ductless glands .	Exocrine glands have ducts .
Secretion	Their secretions are called hormones .	Their secretions include sweat, enzymes, mucus, and sebum.
Route of Secretion	Secretory products are released directly into the bloodstream , eventually reaching the target organ.	Secretory products are released to an internal organ or the external surface through a duct.
Examples	Thyroid gland, parathyroid gland, pituitary gland, adrenal glands.	Salivary glands, liver (exocrine for bile secretion), Brunner's glands, oesophageal glands.

7. HUMAN ENDOCRINE SYSTEM

- The endocrine glands and hormone-producing diffused tissues/cells located in different parts of our body constitute the endocrine system.
- The **pituitary, pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads (testis in males and ovary in females)** are the organised endocrine glands in our body.



1. Hormones

- **Hormones** are chemical messengers secreted directly into the bloodstream by **endocrine glands or specialized endocrine cells**.
- They travel through the blood to **target cells/organs**, where they produce specific responses.
- Hormones act by binding to **specific receptors** on target cells, ensuring **selective action**.
- Excess hormone levels cause a decrease in receptor numbers (**downregulation**), reducing cellular sensitivity.
- Low hormone levels increase receptor numbers (**upregulation**), enhancing cellular sensitivity.
- The hormone transport, interaction with their targets, and removal from the body are dependent on their chemical structure.
- There are two major types of hormones: **lipid-soluble and water-soluble**.

GLAND	HORMONE	TARGET ORGAN/TISSUES	FUNCTION
Pineal Gland	Melatonin	Multiple tissues	Biological clock
Pituitary gland	FSH/LH TSH ADH Growth Hormone Oxytocin Prolactin	Ovaries Thyroid gland Kidneys Multiple tissues Uterus Breast tissue	Menstrual cycle/ovulation Thyroid hormone Osmoregulation Growth & cell division Birth contractions Milk production
Thyroid Gland	Thyroxine	Most body cells	Metabolism regulation
Adrenal Glands	Adrenaline Cortisol	Multiple organs Multiple organs	Fight Or Flight response Anti-Stress
Pancreas	Insulin/Glucagon	Liver, muscles, adipose tissues	Regulate blood sugar levels
Ovaries	Estrogen/Progesterone	Uterus	Menstrual Cycle/Pregnancy
Testes	Testosterone	Many	Male secondary sexual Characteristics

1.1. Hypothalamus

- The hypothalamus is a tiny gland located toward the base of the brain and lies above the pituitary gland in the forebrain. It regulates body temperature and is also known as the **“thermostat”** of the body.
- The **hypothalamus** also directs the **pituitary gland (master gland)** to produce and release hormones and hence is often known as the **master of the master gland**.

1.2. The Pituitary Gland

- A **small, pea-sized endocrine gland** located at the base of the brain **below the hypothalamus**.
- Connected to the hypothalamus through a stalk of blood vessels and nerves (the pituitary stalk) and forms a hypothalamus-pituitary complex that serves as the brain's central command centre to control vital bodily functions.
- The gland referred to as the "**Master Gland**" is so named because it instructs other glands within the endocrine system to release hormones.
- Responsible for the production of several hormones, including **growth hormone (GH)**, **prolactin (PRL)**, **thyroid-stimulating hormone (TSH)**, **adrenocorticotrophic hormone (ACTH)**, **luteinizing hormone (LH)**, **follicle-stimulating hormone (FSH)**, **melanocyte-stimulating hormone (MSH)**, **oxytocin**, and **vasopressin**.
- **LH** and **FSH** (gonadotrophins) regulate gonadal activities.
- In males, **LH** stimulates androgen secretion from the testes, while **FSH**, along with androgens, regulates **spermatogenesis**.
- In females, **LH** triggers ovulation and maintains the **corpus luteum**, while **FSH** promotes the growth and maturation of ovarian follicles.
- **ACTH (Adrenocorticotrophic Hormone)** stimulates the secretion of **glucocorticoids** from the adrenal cortex.
- **TSH (Thyroid-Stimulating Hormone)** stimulates the synthesis and release of thyroid hormones.
- **Prolactin** regulates mammary gland growth and milk production.
- Excess **Growth Hormone (GH/Somatotropin)** causes **gigantism**, while its deficiency causes **pituitary dwarfism**.
- **Oxytocin**, synthesized in the hypothalamus and released by the pituitary gland, stimulates smooth muscle contraction, uterine contractions during childbirth, and milk ejection from mammary glands. It is also known as the **love hormone**, **bonding hormone**, or **birth hormone (Pitocin)**. Although oxytocin influences sexual motivation, it is **not a sex hormone**.
- **Vasopressin** acts mainly on the **kidney and stimulates the reabsorption of water** and electrolytes by the distal tubules, thereby reducing loss of water through urine (diuresis). Hence, it is also called Antidiuretic Hormone (ADH).

1.3. The Pineal Gland (Third Eye Gland – Smallest Endocrine Gland)

- The pineal gland is located on the **dorsal side of the forebrain**.
- The pineal gland secretes a hormone known as **melatonin**, which is crucial for the regulation of a **24-hour circadian (diurnal) rhythm of our body and also assists in controlling sexual maturation**.
- It helps in maintaining the normal rhythms of the sleep-wake cycle and body temperature.

1.4. Thyroid Gland

- The **thyroid gland** is located in the front of the neck and is **butterfly-shaped**.
- It secretes **Triiodothyronine (T3)**, **Thyroxine (T4)**. Additionally, the thyroid gland generates **Calcitonin**, which is crucial for maintaining **calcium homeostasis**.
- **T3 and T4** regulate metabolism and affect nearly all body tissues.
- The thyroid gland consists of **follicles and stromal tissues**.
- **Thyroxine (T4)** influences behaviour and is often called the "**Temperament Gland**." **Iodine** is essential for thyroid hormone synthesis; its deficiency can cause **hypothyroidism** and **goitre**.
- Thyroid hormones regulate the **basal metabolic rate (BMR)**, support **RBC formation**, and control the metabolism of **carbohydrates, proteins, and fats**.

Disorders:

1. **Cretinism:** Physical and mental growth decreases in child.
2. **Myxoedema:** Puffy and swollen face and thickened and dry skin.

- 3. Hashimoto's Disease:** The body's immune system attacks the thyroid gland, also known as the **suicide of the thyroid gland**.

1.5. Parathyroid Gland

- Humans have **four parathyroid glands** located on the posterior surface of the thyroid gland.
- They secrete **Parathyroid Hormone (PTH)**, regulated by blood calcium levels.
- PTH** acts on bones and stimulates **bone resorption (demineralisation)**.

1.6. Adrenal Gland

- A pair of **adrenal glands** is located above each kidney.
- Each gland consists of an inner **adrenal medulla** and an outer **adrenal cortex**.
- The adrenal medulla secretes **adrenaline (epinephrine)** and **noradrenaline (norepinephrine)**, collectively called **catecholamines**. These hormones are released during stress and emergencies and are known as **fight-or-flight** or **emergency hormones**.
- They increase **heart rate, cardiac output, breathing rate, blood glucose levels**, and promote the breakdown of **glycogen, fats, and proteins**.
- The adrenal gland also secretes **cortisol**, known as the **stress hormone**.
- Other hormone secreted by the Adrenal gland is **Aldosterone**, which helps retain **sodium and water**, maintaining blood volume; hence, it is called the **life-saving hormone**.

Disorders:

- Cushing's Syndrome:** Excess cortisol levels.
- Addison's Disease:** Deficiency of cortisol or aldosterone causing weakness, fatigue, low blood pressure, and nausea.

1.7. Pancreas

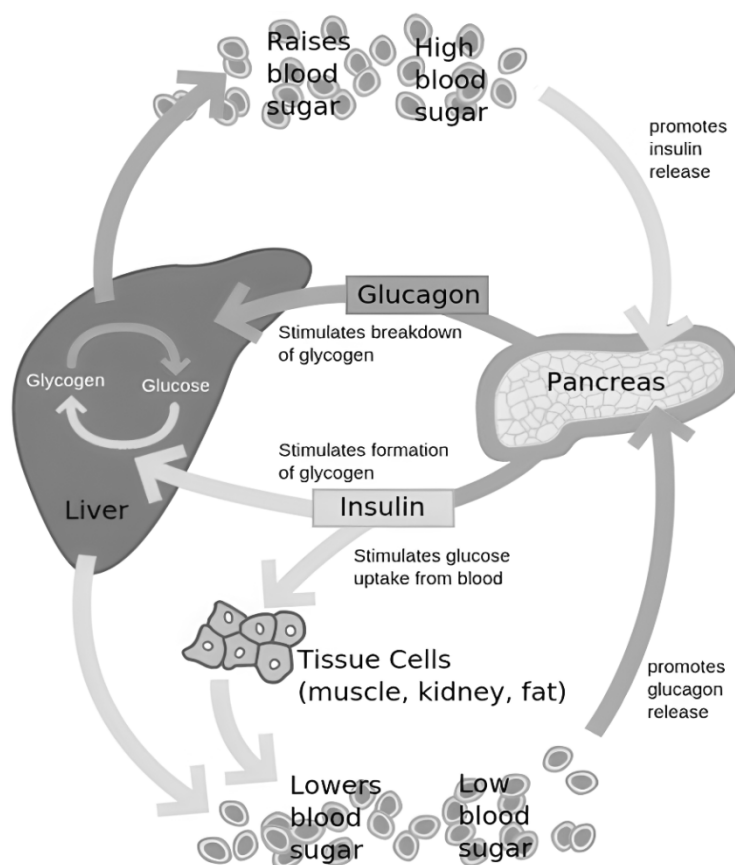
- The **pancreas** is a **mixed (heterocrine) gland**, functioning as both an **exocrine** and **endocrine** gland.
- The endocrine part consists of **Islets of Langerhans**, numbering about **1–2 million** in humans.
- The two main cell types are **α -cells (alpha cells)** and **β -cells (beta cells)**. **α -cells** secrete **glucagon**, while **β -cells** secrete **insulin**.

1) Glucagon

- Glucagon** is a **peptide hormone** that helps maintain normal blood glucose levels.
- It acts mainly on the **liver (hepatocytes)** and stimulates **glycogenolysis**, thereby increasing blood glucose levels (**hyperglycaemic effect**).

2) Insulin

- Insulin** is a **peptide hormone** consisting of **51 amino acids**, secreted by the **β -cells of the Islets of Langerhans**.



- It regulates the **blood glucose level** and is released when blood glucose rises, such as after a meal.
- When blood glucose falls, insulin secretion decreases, and the liver releases glucose into the blood.
- Insulin regulates **carbohydrate metabolism** by controlling glucose utilization.
- The production of insulin is an **endocrine function** of the pancreas.
- **Hyposecretion of insulin** causes **diabetes mellitus**.
- The roots of **Dahlia** are a source of **inulin (not insulin)**, which has anti-diabetic properties and is used by diabetic patients.

1.8. Testis

- A pair of **testes** is present in the **scrotal sac** outside the abdomen.
- Testes function as both **primary male sex organs** and **endocrine glands**.
- **Leydig cells** secrete **androgens**, mainly **testosterone**.
- Testosterone regulates the development, maturation, and functions of male accessory sex organs.
- It promotes **muscle growth, facial and axillary hair growth, aggressiveness, and deepening of the voice**.
- **Androgens** play an important role in **spermatogenesis** and influence **male sexual behaviour** through their action on the central nervous system.

1.9. Ovary

- Females have a pair of **ovaries** located in the abdomen.
- The ovary is the **primary female sex organ**, producing an **ovum** during each menstrual cycle.
- It also secretes the steroid hormones **estrogen** and **progesterone**.

A. Estrogen

- **Estrogen** plays an important role in female sexual and reproductive development.
- Pituitary hormones stimulate estrogen production at puberty.
- The three types of estrogen are **estrone, estradiol, and estriol**.
- Estrogen regulates the female reproductive system and secondary sexual characteristics such as **breast development, wider hips, pubic and axillary hair, and vaginal maturation**.

B. Progesterone

- After ovulation, the ruptured **Graafian follicle** forms the **corpus luteum**, which secretes **progesterone**.
- During pregnancy, the **placenta** also produces progesterone.
- Progesterone prepares the **uterine lining** for implantation, supports **pregnancy**, and suppresses estrogen secretion after ovulation.
- It also stimulates the formation of **alveoli** in the mammary glands and promotes **milk secretion**.
- Both **testes and ovaries** produce small amounts of hormones of the opposite sex; for example, females also secrete a small amount of **testosterone**.

BCW Facts:

- **Globular proteins** exhibit a spherical configuration created by coiled **polypeptide chains**. Typically, these proteins are soluble in water. **Insulin and albumins** serve as common examples of globular proteins.
- In the early 1920s, **Frederick Banting** and **Charles Best** discovered and isolated **insulin** under the leadership of John Macleod at the University of Toronto. With assistance from James Collip, insulin was purified, rendering it available for the effective treatment of diabetes. In 1923, the **Nobel Prize in Physiology or Medicine** was conferred upon **Frederick Banting** and **John Macleod** for their **discovery of insulin**.

8- CIRCULATORY SYSTEM

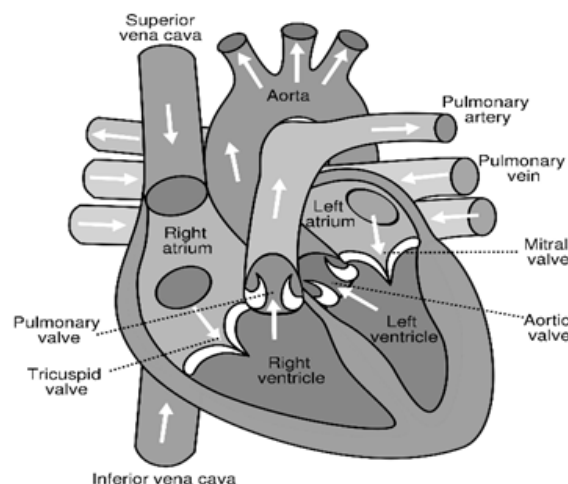
1. Circulatory System

- The circulatory system is responsible for transporting essential nutrients and minerals throughout the body and for removing metabolic waste products.
- It consists of a **network of arteries, veins, and capillaries, with the heart pumping blood through it.**

1.1 Types of Circulatory System

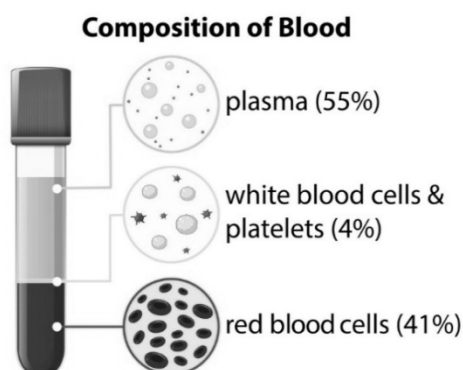
Depending on the circulation method, the circulatory system can be classified as either open or closed.

- Open circulatory system:** In this system, blood does not circulate within closed vessels; instead, it flows through various parts of the body cavity, remaining mixed with bodily fluids. Organisms such as **prawns and insects** possess an **open circulatory system**.
- Closed circulatory system:** Blood circulates within well-defined, tube-like vessels. A sufficiently high pressure is maintained within this system. The closed system is more efficient compared to the open type. **The closed circulatory system is present in all vertebrates.**

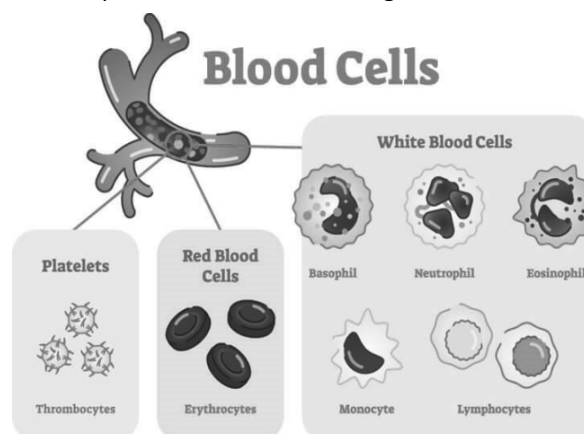


2. Blood

- Blood serves to transport food, oxygen, nutrients, and waste products throughout our bodies. It is classified as a fluid **connective tissue**.
- Blood is composed of a liquid medium known as plasma, in which the cells are suspended. Plasma carries food, carbon dioxide, and nitrogenous wastes in a dissolved state.
- Oxygen is delivered by the red blood cells.** Additionally, various other substances, such as salts, are also transported by the blood.
- A pumping organ is required to circulate blood throughout the body, along with a network of vessels to reach all tissues, and a system to ensure that this network can be repaired if it becomes damaged.



Key



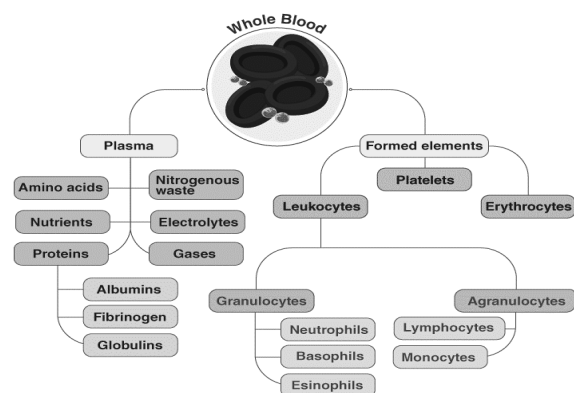
Points about Blood:

- It consists of a fluid matrix, **plasma**, and formed elements like **RBCs & WBCs**.
- Blood have fluid medium called plasma to transport food, CO₂, Nitrogenous waste, etc.

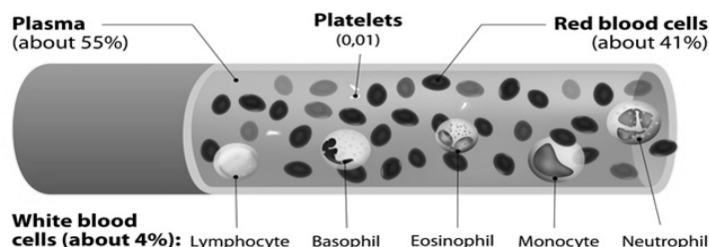
- The female body has an average 4.5L & 5.5L in male.
- Produced from the **stem cell of bone marrow**, which is hematopoietic cell.
- Oxygen is transported by RBC corpuscles.
- PH is slightly alkaline, which is 7.4, and it is 3-5 times more viscous than water.
- Transports food, oxygen, and metabolic materials.

2.1 Functions Of Blood

- **Transportation:** respiration, nutrient carrier from gastro intestinal track, transportation of hormones from endocrine glands, and transports metabolic wastes.
- **Regulation:** Regulates pH, adjusts and maintains body temperature, maintain water content of cells
- **Protection:** WBC protects against diseases by phagocytosis, reservoir for substance like water, electrolytes, etc., and performs haemostasis.



2.2 The Elements of Blood



3. Erythrocytes Or Red Blood Cells (RBCs)

- **RBCs (Red Blood Cells)** are the most abundant cells in the blood.
- A healthy adult man has, on an average, **5 million to 5.5 million of RBCs mm^{-3}** of blood. They are produced in the **red bone marrow**.
- In most mammals, RBCs are **biconcave** and **lack a nucleus**.
- RBCs contain **haemoglobin (Hb)**, an iron-rich protein complex; each haemoglobin molecule contains **four iron atoms** and transports **oxygen**.
- RBCs have an average lifespan of **120 days** and are destroyed mainly in the **spleen** (the "graveyard of RBCs").
- During **hypoxia** (low oxygen levels), the kidneys secrete a hormone **erythropoietin**, which stimulates RBC production.
- **100 ml of blood** typically contains about **12–16 grams of haemoglobin**.

4. Leukocytes Or White Blood Cells (WBCs)

- **WBCs** lack haemoglobin, are **nucleated**, and occur in smaller numbers (averages $6000-8000 \text{ mm}^{-3}$ of blood).
- They are classified into **Granulocytes (Neutrophils, Eosinophils, Basophils)** and **Agranulocytes (Monocytes, Lymphocytes)**.

Granulocytes

- **Neutrophils:** Phagocytic cells that destroy foreign organisms; **most abundant WBCs** (60-65% of total WBCs).
- **Basophils:** Secrete **histamine, serotonin, and heparin**; involved in **inflammatory reactions**; **least abundant WBCs** (0.5-1% of total WBCs).
- **Eosinophils:** Defend against infections and are associated with **allergic reactions**.

Agranulocytes

- **Monocytes:** Phagocytic cells that remove dead cells and protect against blood-borne pathogens.
- **Lymphocytes (20–25%):** Produced in the bone marrow and are of two types—**B cells** and **T cells**; both are involved in immune responses.
 - **T cells:** Mature in the **thymus** and help eliminate pathogens.
 - **B cells:** Mature in the **bone marrow** and produce **antibodies** in response to antigens; on exposure to antigens, they form **plasma cells** and **memory B cells**.

5. Plasma

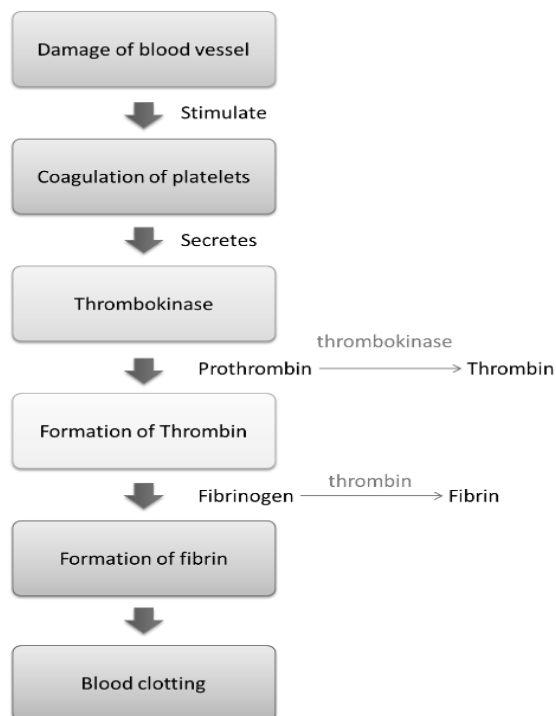
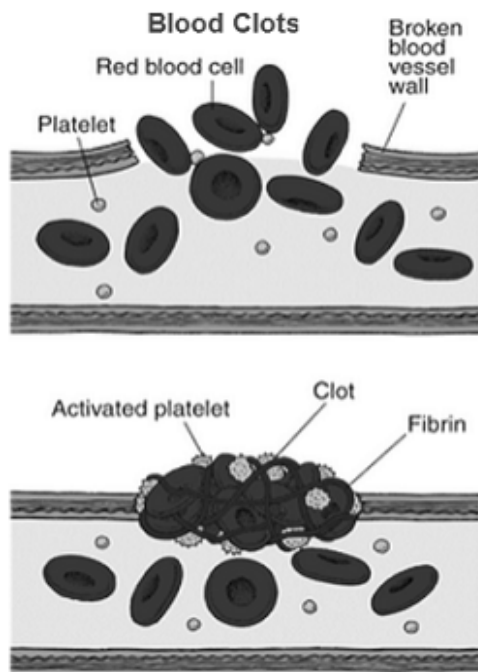
- **Plasma** is a **straw-coloured, viscous fluid** that constitutes about **55% of blood**.
- It contains **90–92% water** and **6–8% proteins**, mainly **fibrinogen, globulins, and albumins**. **Fibrinogens** are essential for **blood clotting (coagulation)**, **Globulins** are primarily involved in the **body's defence mechanisms** and **Albumins** help maintain **osmotic balance**.
- Additionally, plasma **contains trace amounts of minerals** such as **sodium, calcium, chlorine, magnesium, and hydrogen carbonate**. Glucose, amino acids, lipids, and other substances are also present in plasma as they circulate throughout the body.
- **Coagulation factors**, which are essential for blood clotting, are present in plasma in an **inactive state**. Plasma that lacks these clotting factors is referred to as **serum**.

6. Platelets

- Platelets, also known as **thrombocytes**, are cell fragments derived from **megakaryocytes** in the bone marrow.
- Their normal count ranges from **1,50,000 to 3,50,000 per mm^3 of blood**.
- Platelets release substances involved in **blood clotting (coagulation)**.
- A **low platelet count** can impair clotting and cause **excessive bleeding**.

7. Blood Clotting

- **Blood clotting (coagulation)** occurs after injury or trauma to prevent excessive blood loss.
- A **dark reddish-brown scab** forms at the injury site, consisting mainly of **fibrin threads** that trap blood cells and damaged tissues.
- **Fibrin** is formed when **fibrinogen** is converted into fibrin by the enzyme **thrombin**.
- **Thrombin** is produced from inactive **prothrombin**. This conversion requires the enzyme complex **thrombokinase** and occurs through a **cascade of enzymatic reactions**.
- Injury stimulates **platelets** and damaged tissues to release factors that initiate clotting.
- **Calcium ions** are essential for the coagulation process.



8. Lymph

- As blood flows through **tissue capillaries**, some water and soluble substances leak out to form **tissue (interstitial) fluid**. Tissue fluid facilitates the exchange of **nutrients, gases, and wastes** between blood and cells. It has a **mineral composition similar to plasma**.
- Tissue fluid is collected by the **lymphatic system** and returned to the veins. The fluid in the lymphatic system is called **lymph**, a colourless fluid rich in **lymphocytes**.
- Lymph supports **immunity**, transports **nutrients and hormones**, and absorbs **fats through lacteals** in the intestinal villi.

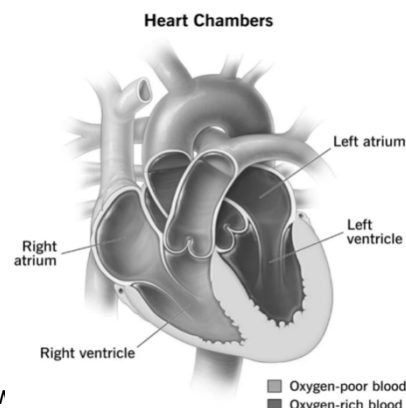
Bone marrow

It is semi-solid tissue found within the spongy (also known as cancellous) portions of bones. It is composed of adipose tissue, areolar tissue, and blood. There are two types of bone marrow.

- Red bone marrow** - It produces blood cells. Red bone marrow contains blood stem cells that convert into red blood cells, white blood cells, or platelets.
- Yellow bone marrow** - It stores fat and produces red blood cells during life-threatening situations.

9. The Heart

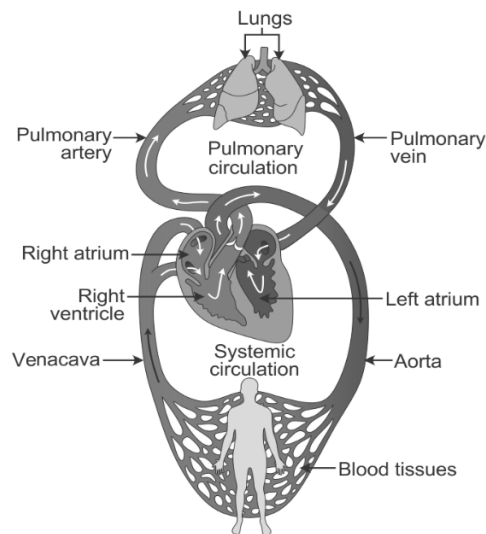
- The heart is a muscular organ, about the size of a fist, located in the chest cavity.
- It pumps blood throughout the body through blood vessels, a process known as **circulation**.
- It supplies oxygen and nutrients to tissues and removes carbon dioxide and other metabolic wastes.



- The human heart weighs approximately **250–300 g**.
- The **heart, blood, and blood vessels** together form the **cardiovascular system**.
- The membrane enclosing the heart is called the **pericardium**.

10. Chambers Of Heart

- **Right atrium/auricles:** Deoxygenated blood comes from the body to the upper chamber, the right atrium, through the vena cava, as it relaxes or inflates.
- **Right ventricle:** As the right atrium contracts, the corresponding lower chamber, the right ventricle, inflates, and blood is transferred to the right ventricle through the Tricuspid valve, which in turn pumps it to the lungs for oxygenation through the Pulmonary artery.
- **Left atrium/auricles:** Oxygen-rich blood from the lungs comes to the thin-walled upper chamber of the heart. It relaxes while collecting this blood and contracts while blood is transferred to the next chamber (the left ventricle)
- **Left ventricle:** It contracts while collecting blood transferred from left atrium. The muscular left ventricle contracts so the blood is pumped out to the body.



11. The Tube - Blood Vessels

- The tubes transporting blood are called Blood Vessels. The wall of a blood vessel has three layers: tunica externa, tunica media, and tunica interna. There are **3 kinds of blood vessels**: (i) Artery (ii) Capillary and (iii) Vein.

Arteries

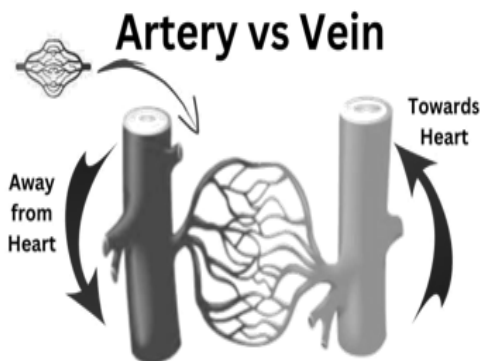
- Arteries carry blood **away from the heart** to various body organs.
- They usually transport **oxygenated blood**, except the **pulmonary artery**, which carries deoxygenated blood to the lungs.
- Their walls are **thick, muscular, and elastic** to withstand high blood pressure.
- The **aorta** is the largest artery in the body.

Veins

- Veins carry blood **towards the heart** from different body parts.
- They usually transport **deoxygenated blood**, except the **pulmonary veins**, which carry oxygenated blood from the lungs.
- Their walls are thinner and contain **valves** that prevent backflow of blood.
- The **vena cava** (superior and inferior vena cava) is the largest vein in the body.

Capillaries

- Arteries branch into smaller vessels and finally into **capillaries**.
- Capillaries are the **smallest blood vessels** and have walls only **one cell thick**.
- Exchange of oxygen, nutrients, carbon dioxide, and wastes occurs through capillary walls.
- Capillaries unite to form **veins**, which return blood to the heart.



BCW Fact: An **electroencephalogram (EEG)** is carried out to record the electrical activity in the brain using metal discs (electrodes) attached to the scalp. EEG helps to diagnose sleep disorders, epilepsy and certain neurological conditions like encephalopathy. An **Electrocardiogram (ECG)** is used to record the electrical activity of the heart produced by the excitation of the cardiac muscles. ECG uses electrodes to record electrical activity.

12. Circulation Of Blood In Various Organisms

- Separation of the right and left sides of the heart prevents the mixing of oxygenated and deoxygenated blood, ensuring an efficient oxygen supply to the body. This is especially important in **birds and mammals**, which have high metabolic rates and maintain a constant body temperature. They possess a **four-chambered heart**.
- **Amphibians** and most **reptiles** (except crocodiles, which have a four-chambered heart) possess a **three-chambered heart**. Some mixing of oxygenated and deoxygenated blood occurs, which is adequate for their lower metabolic demands compared to birds and mammals.
- Amphibians can also absorb oxygen through their **moist skin (cutaneous respiration)**, making a completely separated circulatory system less essential.
- **Fish** have a **two-chambered heart**. Blood is pumped from the heart to the gills for oxygenation and then directly to the rest of the body before returning to the heart.
- In fish, blood passes through the heart **once** during each complete circuit (**single circulation**). In other vertebrates, blood passes through the heart **twice** during each circuit (**double circulation**).

13. Blood Pressure

- Blood pressure is the **force exerted by blood against the walls of blood vessels**, especially arteries. It is higher in arteries than in veins.
- During **ventricular systole (contraction)**, arterial pressure is highest and is called **systolic pressure**.
- During **ventricular diastole (relaxation)**, arterial pressure is lowest and is called **diastolic pressure**.
- A normal adult blood pressure is approximately **120 mm Hg** (systolic) and **80mm Hg** (diastolic).
- Blood pressure is measured using a **sphygmomanometer**.
- **mm Hg** stands for **millimetres of mercury**, the unit of pressure.
- **Hypertension (high blood pressure)** is a condition of persistently elevated blood pressure and may damage blood vessels and increase the risk of **arterial rupture, stroke, or internal bleeding**.

14. Blood Group

- **Karl Landsteiner**, an Austrian scientist, discovered the ABO blood group system in the year 1900. An antigen is a substance that stimulates the production of antibodies.
- A person's blood type is classified by the presence or absence of A and B antigens and Rh factor on the surface of the blood cells.
- The **antigens A and B** are chemicals that can induce an immune response during blood transfusion.
- Similarly, the plasma of different individuals contains two natural antibodies or proteins produced in response to antigens. Therefore, the blood of a donor has to be carefully matched with the blood of a recipient before any blood transfusion to avoid severe problems of clumping (destruction of RBC).

The donor's compatibility is also shown in the table given below.

Blood Group	Antigens On RBCs	Antibodies In Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	AB	nil	AB, A, B, O

O	NIL	anti-A, B	O
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- Group 'O' blood can be donated to persons with any other blood group and hence 'O' group individuals are called 'universal donors'.
- Persons in the 'AB' group can accept blood from persons with AB as well as the other groups of blood. Therefore, such persons are called 'universal recipients'.

15. Rh Factor

- The Rh (Rhesus) factor is the positive (+) or negative (–) marker in blood groups, representing an inherited protein present on RBCs. It was discovered in 1940 by Landsteiner and Wiener. Rh incompatibility can lead to destruction of red blood cells.
- **Rh status:**
 - Rh+ blood: RBCs contain Rh protein.
 - Rh– blood: RBCs lack Rh protein.
- **Antibody response:**
 - Rh+ individuals do not produce anti-Rh antibodies.
 - Rh– individuals produce anti-Rh antibodies if exposed to Rh+ blood.
- **Transfusion compatibility:**
 - Rh+ individuals can receive Rh+ and Rh– blood.
 - Rh– individuals can receive only Rh– blood.

Erythroblastosis foetalis

- It is a potentially dangerous condition that occurs during the development of the foetus. The condition occurs when a component of blood called the Rh factor is incompatible between the pregnant woman and the foetus.
- It can cause jaundice and other, more severe complications, including possible heart failure. **Erythroblastosis foetalis** is also known as **Hemolytic disease of the foetus and newborn (HDFN)**.

Different colours of Blood found in various organisms

Colour Of Blood	Pigment Present	Found In
Red	Haemoglobin	Humans and the majority of other vertebrates
Blue	Hemocyanin	Spiders, crustaceans, octopus, some molluscs, octopuses and squid, crayfish
Green	Chlorocruorin	Some segmented worms, polychaete groups, some leeches, and some marine worms,
Violet	Hemerythrin	Marine worms including peanut worms, brachiopods
Yellow	Vanabin	Beetles, sea squirts, sea cucumber
Colourless/white	No respiratory pigment present	Cockroach

16. Cardiac Cycle

- The **cardiac cycle** is the sequence of events during one complete heartbeat, from the beginning of one beat to the start of the next.
- It consists of **systole (contraction), diastole (relaxation), and a brief pause**.
- A healthy adult heart beats about **72 times per minute**, corresponding to **72 cardiac cycles per minute**.

- One cardiac cycle involves the complete contraction and relaxation of the **atria and ventricles** and lasts about **0.8 seconds**.

The different phases of the cardiac cycle involve:

1. Atrial diastole
2. Atrial systole
3. Isovolumic contraction
4. Ventricular ejection
5. Isovolumic relaxation
6. Ventricular filling

17. Sinoatrial Node

- The **sinoatrial (SA) node** or **sinus node** is an oval-shaped region of specialized cardiac muscle in the upper back wall of the **right atrium**, made of **pacemaker cells**.
- It is about **15 mm long, 3 mm wide, and 1 mm thick**, and is located directly below and to the side of the **superior vena cava**.
- The SA node generates **electrical impulses (cardiac action potentials)** that travel through the heart's conduction system and initiate **cardiac contraction**.

BCW Facts:

- Lymphoid organs** are sites for the origin, maturation, and proliferation of **lymphocytes**. Primary lymphoid organs—**bone marrow and thymus**—produce antigen-sensitive lymphocytes. Secondary lymphoid organs—like **spleen, lymph nodes, tonsils, Peyer's patches, and appendix**—facilitate lymphocyte–antigen interaction and effector cell formation.
- Tachycardia** is a form of heart arrhythmia characterized by a heart rate exceeding 100 beats per minute (bpm).

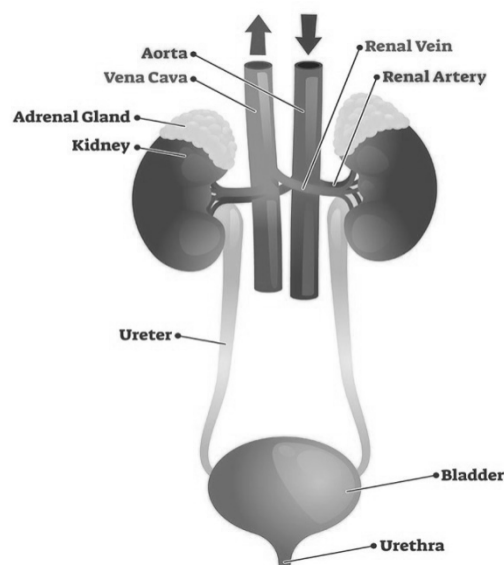
9- HUMAN EXCRETORY SYSTEM

Excretion In Humans

- Excretion** is the process of removing metabolic waste products from the body.
- In humans, various life processes such as **respiration, digestion, and circulation** produce waste products such as **carbon dioxide, water, urea, ammonia, and uric acid**.
- In lower organisms, excretion mainly occurs through **diffusion**. Because simple diffusion is insufficient, humans need a **specialized excretory system** to efficiently eliminate these wastes and maintain internal balance.

Human Excretory System

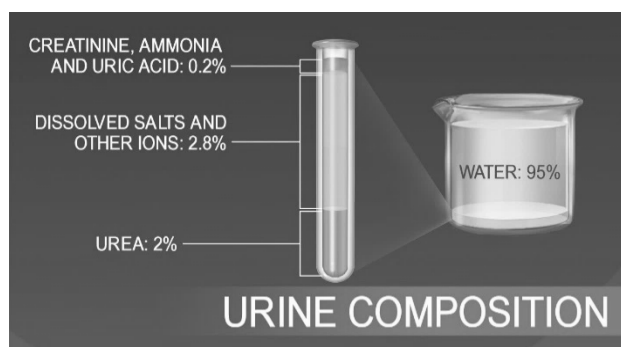
- Consists of a **pair of kidneys, one pair of ureters, a urinary bladder, and a urethra**.
- Kidneys are the major excretory organs; they are **bean-shaped** and located near the dorsal inner wall of the abdominal cavity.
- Each kidney in human beings is about **10–12 cm long** and weighs **120–170 g**.
- The size of the **left kidney is larger than the right kidney**, as the liver pushes the right kidney.



- Each kidney has two distinct zones: an **outer cortex** and an **inner medulla**.
- Each kidney contains about **one million nephrons**, the **structural and functional units** responsible for blood filtration.
- **Nephrons (tube-like structures)** are the structural and functional units of kidneys. Each nephron has two main parts – the **glomerulus** and the **renal tubule**.
 - **The glomerulus** is a cluster of **capillaries formed by the afferent arteriole**, a fine branch of the renal artery. Blood from the glomerulus is carried away by an efferent arteriole.
 - **The renal tubule** begins with a double-walled cup-like structure called **Bowman's capsule**, which encloses the glomerulus. The glomerulus, along with Bowman's capsule, is known as the **renal corpuscle**. The tubule continues further to form a highly coiled network called the proximal convoluted tubule (PCT).
- Kidney capillaries **filter blood** to remove wastes. Useful substances such as **glucose, amino acids, salts, and required water** are **reabsorbed** into the bloodstream.
- Excess water and **nitrogenous wastes** are converted into **urine**. Urine passes through the **ureters** to the **urinary bladder** for storage.
- The **Central Nervous System** is responsible for controlling the urinary bladder. The brain signals the bladder to contract, and urine is excreted through the **urethra**.

1. Urine

- **Human urine** consists mainly of **water (95%)**, **urea (2%)**, **creatinine (0.1%)**, and **uric acid (0.03%)**.
- It also contains small amounts of **chloride, sodium, potassium, sulphate, ammonium, phosphate**, and other ions.
- **Proteins are present only in trace amounts**, much lower than their concentration in **blood plasma**.



Urea ($\text{CH}_4\text{N}_2\text{O}$)

Wastes are produced due to metabolic activities in our body. Urea is produced when protein-rich food is broken down into amino acids.

- When **amino acids are metabolised in the liver**, they produce **free ammonia**, which is **highly toxic** to the body. This is further converted into **non-toxic urea**, which the body can easily eliminate through the kidneys in urine.

Protein → Amino Acids → Ammonia → Urea

(Highly toxic) → (Less toxic)

- **Creatinine** is a waste product generated by muscles and is part of regular & everyday activities.

1.1 Urine Formation

- **Urea** is excreted mainly through **urine**, and kidneys are the main excretory organs.
- It is produced in the **nephron, which is the structural and functional unit of the kidney**.
- Urine formation in the body is mainly carried out in three phases, namely: Glomerular filtration, reabsorption, and secretion.

a. Glomerular Filtration

- First step in urine formation. Excess fluid and waste products from the kidney are filtered out of the blood into **Bowman's capsule** to be eliminated from the body.
- The amount of filtrate produced by the kidneys every minute is known as **Glomerular Filtration Rate**. On average, **1100 – 1200 ml of blood is filtered by the kidneys per minute**.

b. Tubular Reabsorption (selective)

- This involves the **reabsorption of essential ions and compounds** such as water, glucose, amino acids, and sodium ions. As the urine passes through the tube, this takes place.
- The amount of water that is reabsorbed depends upon the amount of extra water in the body as well as the amount of dissolved waste to be expelled.

c. Secretion/Tubular Secretion

- **Tubular secretion** involves secretion of **K⁺, H⁺, and NH₃** into the tubule to maintain **fluid and acid-base balance**.
- Urine formed in each kidney flows through the **ureters**, which connect the kidneys to the **urinary bladder**.
- The bladder stores urine until it is expelled through the **urethra**.
- Since the bladder is muscular, its receptors communicate with the brain, allowing the sphincter muscles to relax and discharge it. Normally, in a healthy adult, the initial **Glomerular filtrate in the kidneys is about 180 litres daily**. This is known as **micturition**.
- However, only **1–2 liters/day** of urine are excreted, as around 99% of the filtrate is reabsorbed in the kidney tubules.

BCW Facts:

- **Cystitis** is inflammation of the bladder, usually caused by a bladder infection. It's a common type of urinary tract infection (UTI), particularly in women.
- Reptiles, birds, and terrestrial snails excrete **nitrogenous waste as "Uric Acid" and are called uricotelic animals**.

2. Spleen

- The **spleen** is a large bean-shaped organ **inside the left rib cage just above the stomach**.
- It contains **lymphocytes and phagocytes**.
- The spleen is known as the "**graveyard of RBCs**" because it is the major site where aged or damaged RBCs are destroyed by **splenic macrophages**.
- During RBC breakdown, **haemoglobin** is split into haem and globin.
- Iron released from haem is transported by **transferrin** to the bone marrow for reuse in RBC production.
- **Heme is converted into bilirubin**, a yellow-brown pigment. The liver secretes bile pigments (bilirubin, some biliverdin), which are metabolic wastes of haemoglobin of dead RBCs.
- **Urochrome is eliminated through urine**. Urochrome is the yellow pigment responsible for the pale-yellow colour of urine.

3. Kidney Stone (Renal Calculi)

- Kidney stones are **hard mineral deposits** formed in the urinary tract. Composed of substances like **calcium oxalate, uric acid, struvite, and cystine**.

- Form when urine becomes concentrated, allowing calcium oxalate and other minerals to crystallize and stick together.
- Can occur anywhere from the **kidneys to the bladder** and may cause severe pain.
- Treatment includes:
 - Pain relievers and increased water intake.
 - **Surgical removal** in severe or obstructive cases.

4. Dialysis

- **Dialysis** is an artificial process that performs **blood filtration**. It is used when kidneys fail due to disease or injury, typically when function drops to about **10–15%**.
- Without dialysis, **wastes and salts accumulate in blood**, damaging other organs.
- There are **two types of dialysis: haemodialysis and peritoneal dialysis**.
 - **Haemodialysis** helps remove excess fluids and waste products from the blood when the kidneys fail to function properly.
 - **Peritoneal dialysis** uses the **peritoneal membrane** (lining of the abdominal cavity) to filter blood.

5. Sweating

- The **sweat and sebaceous glands in the skin** can eliminate certain substances through their secretions. Sweat produced by the sweat glands is a watery fluid containing **NaCl, small amounts of urea, lactic acid, etc.**
- Sweat primarily helps in **cooling the body** and also removes some **metabolic wastes**.
- **Sweat glands** are located in the **dermis** and release sweat through ducts opening on the skin surface.
- **Sebaceous glands**, found near hair follicles, secrete **sebum** that lubricates and protects the skin from infection and injury.
- The **dermis** contains sensory nerve endings for **pain, touch, and temperature**, sending signals to the central nervous system (CNS).
- The **skin regulates body temperature** and helps maintain thermal balance.

BCW Facts:

- **Glomerulonephritis** occurs when the blood filters of the kidneys, known as **glomeruli**, become inflamed and scarred. If left untreated, this condition can progress to kidney failure.
- **Lithotripsy** is a **non-invasive procedure used to address kidney stones** that are too large to be expelled through the urinary tract. This procedure utilizes a series of shock waves directed at the stone.
- **Bright's disease** refers to a historical classification of kidney disorders. In cases of Bright's disease, the blood vessels in the kidneys are inflamed, resulting in the excretion of **albumin in the urine**. The disease is now referred to as '**nephritis**'.

10- RESPIRATION

Respiration is a fundamental biochemical process that provides the energy required for life. It supports vital functions such as growth, movement, repair, reproduction, and maintenance of homeostasis. Respiration occurs in all living organisms, from unicellular bacteria to multicellular humans, and is essential for survival.

1. Respiration Process

- Respiration is a metabolic process that occurs in all organisms. It is a biochemical process that takes place within the cells of organisms.
- In this process, **energy (ATP-Adenosine triphosphate)** is produced by the breakdown of glucose, which is further used by cells to perform various functions.
- Every living species, from single-celled organisms to dominant multicellular organisms, performs respiration.

1.1 Sub-processes of Respiration:

a. Breathing

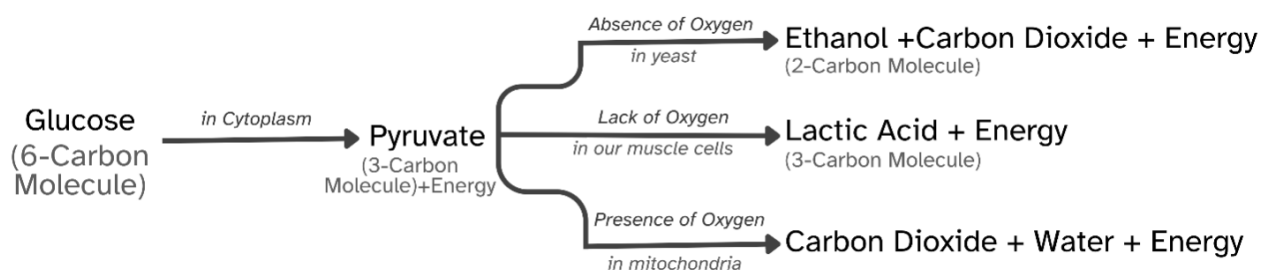
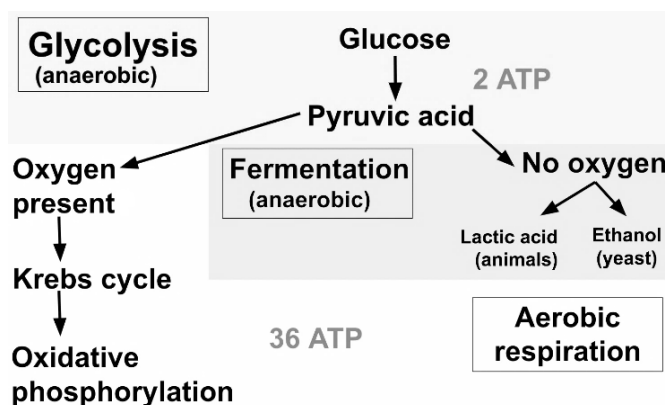
- Breathing is the **process of moving air into and out of the lungs to facilitate gas exchange with the internal environment**. It involves the **intake of oxygen (O_2)** and the **release of carbon dioxide (CO_2)**.

b. Cellular respiration

- Cellular respiration is the breakdown of simple food molecules** in order to release energy inside the cell.

Respiration is of two types:

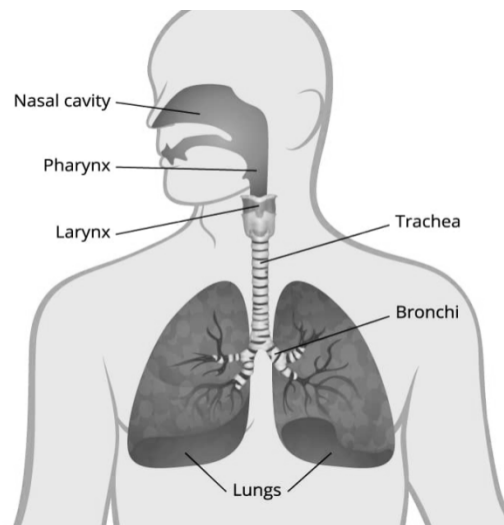
- Aerobic respiration
- Anaerobic respiration



2. Human Respiratory System

- In human beings, air enters the body through the **nostrils**, where **fine hairs and mucus** filter dust and other particles.
- The **nasal chamber** opens into the **pharynx**, a common passage for **air and food**.

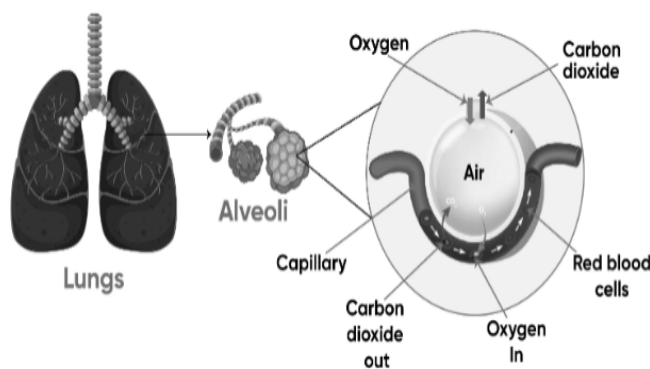
- The pharynx leads to the **larynx (voice box)** and then to the **trachea (windpipe)**.
- During swallowing, the **epiglottis** covers the **glottis**, preventing food from entering the respiratory tract.
- **Cartilaginous rings** in the trachea prevent its collapse and keep the airway open.
- Each bronchus undergoes repeated divisions to form the secondary and tertiary bronchi and bronchioles, ending in very **thin terminal bronchioles**.
- Each **terminal bronchiole** gives rise to several very thin, irregular-walled, and vascularised bag-like structures called **alveoli**.
- The branching network of **bronchi, bronchioles, and alveoli** comprises the **lungs**. In other words, within the lungs, the passage divides into smaller tubes that ultimately terminate in balloon-like structures called **alveoli**.



2.1 Alveoli and Gas Exchange Mechanism

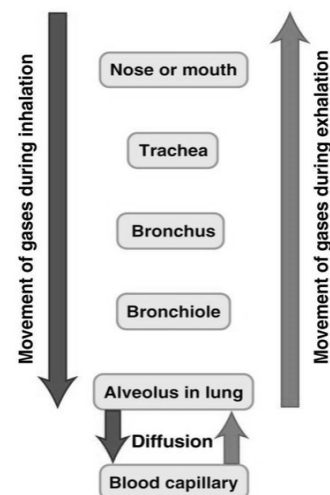
The alveoli provide a surface for gas exchange. The **walls of the alveoli contain an extensive network of blood vessels**.

- The process of breathing is divided into two phases – inhalation (inspiration) and exhalation (expiration).
- **During inhalation**, the **diaphragm contracts** and moves downward. At the same time, the muscles between the ribs contract and pull upward. The **chest cavity expands**, reducing pressure and drawing air into the lungs.
- **During exhalation**, the ribs descend and move inward, while the **diaphragm relaxes**; pressure increases and air is forced out.
- The **blood brings carbon dioxide from the rest of the body for release into the alveoli**, while the oxygen in the alveolar air is taken up by blood in the alveolar blood vessels and transported to all the cells of the body.
- In human beings, the **respiratory pigment is haemoglobin**, which is present in the red blood corpuscles (RBCs).
- The **maximum amount of air that can be exhaled** after a maximal inhalation (inspiration) is called the **vital capacity of the lungs**. It is about 3.5-4.5 litres.
- The total volume of air that the lungs can accommodate at the conclusion of a forced inhalation is referred to as the **total lung capacity**.

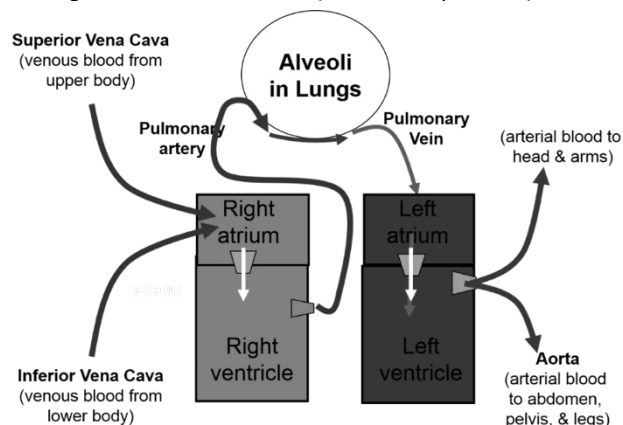


2.2 Respiration involves the following steps:

- **Breathing or pulmonary ventilation**, by which atmospheric air is drawn into the lungs, and CO₂ rich alveolar air is released.
- **Diffusion of gases** (O₂ and CO₂) across the alveolar membrane occurs.
- **Transport of gases** occurs via the bloodstream.



- Diffusion of O_2 and CO_2 occurs **between blood and body tissues**.
- **Utilisation of oxygen by cells for catabolic reactions**, resulting in the release of CO_2 (cellular respiration).



Important Facts About Lungs

- The **right lung is shorter and wider than the left lung**.
- Human lungs are divided into **five Parts**. The **left lung has two lobes (upper and lower)**, and the **right lung has three lobes (upper, middle, and lower)**.
- The two lungs together contain **about 300–500 million alveoli**.

3. Cellular Respiration

3.1 Aerobic Respiration

- It takes place in the presence of oxygen to produce energy. It occurs continuously in the cells of **plants and animals** to meet energy requirements.

This process can be explained with the help of the chemical equation:



- About **2900 kJ of energy** is released from the complete oxidation of one glucose molecule.
- This energy is utilized for the synthesis of **ATP (Adenosine Triphosphate)**, the cell's energy currency.

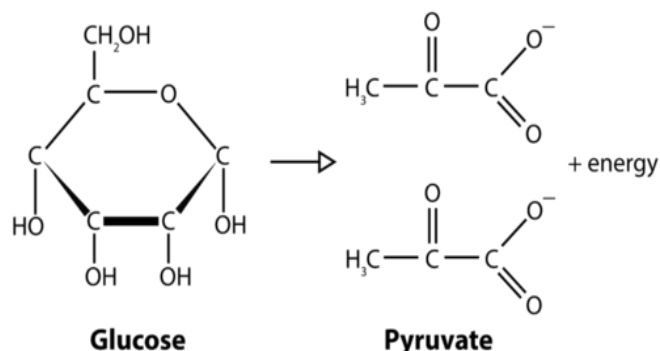
The three stages of aerobic cellular respiration are **glycolysis (an anaerobic process)**, the **Krebs cycle**, and **oxidative phosphorylation**.

Adenosine Triphosphate (ATP)

- The energy released during cellular respiration is immediately used to synthesise a molecule called ATP, which fuels all other cellular activities. Hence, ATP is also known as **"energy currency of the cell"**.
- First discovered in the year 1929 by German chemist Karl Lohmann. In 1948, **Scottish biochemist Alexander Todd** first synthesized the ATP molecule.
- When the body's cells need **energy to perform** metabolic activities, they use ATP and break it down to obtain the necessary energy.
- ATP is mainly synthesized in the mitochondria in the cellular respiratory process through the oxidation of **respiratory substrates** such as carbohydrates, lipids, proteins, etc.
- Their oxidation results in energy production, which is stored in ATP.

Steps involved in respiration:**1. Glycolysis:**

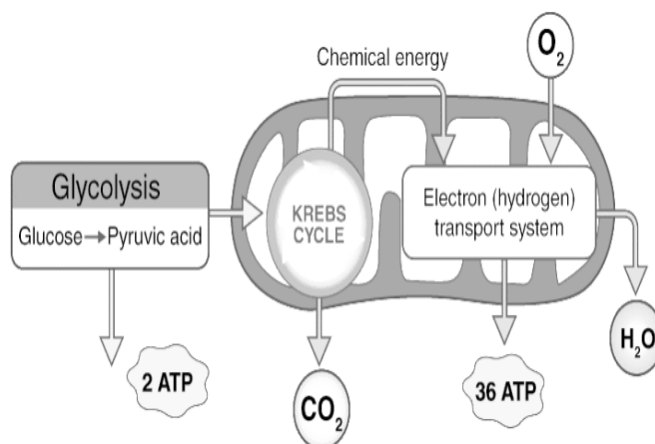
- It is the breakdown of glucose into pyruvate, and this step takes place in the **cytoplasm**.
- Also known as the **EMP pathway**, due to its discovery by three scientists, **Embden, Meyerhof, and Parnas**.
- In this process, **one glucose molecule (6-carbon)** is converted into **two molecules of pyruvic acid (3-carbon each)**.

**2. Fate of Pyruvic Acid:** In the **mitochondria**,

further breakdown of **pyruvic acid** takes place, and the molecules formed depend on the type of respiration in a particular organism.

3. The Krebs cycle:

- It is a series of redox reactions that begins with acetyl-CoA (**acetyl coenzyme A**).
- These reactions take place in the **matrix of the mitochondria** of eukaryotic cells. In prokaryotic cells, it takes place in the **cytoplasm**.
- For each round of the cycle, two carbons enter the reaction in the form of acetyl-CoA.
- The cycle produces two molecules of carbon dioxide. The reactions generate **three molecules of NADH and one molecule of FADH₂**.
- The protons are transported across the membrane using the energy harvested from NADH and FADH₂. When these protons pass through **ATP synthase**, they drive the formation of ATP. Hence, **36 ATP is generated** through this cycle. This is accomplished through a series of electron carriers in the inner mitochondrial membrane, known as the **electron transport system**, which pass electrons, releasing energy to pump protons and drive ATP synthesis in a process called **oxidative phosphorylation**. In this process, O₂ serves as the final electron acceptor, resulting in its reduction to water.

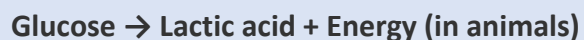


That is why mitochondria are referred to as “**the powerhouses of the cell**.”

NOTE: The **electron transport system** occurs in the **inner mitochondrial membrane**, specifically in its **folded extensions called cristae**.

3.2 Anaerobic Respiration

- It takes place in the **absence of oxygen** to produce energy. Pyruvic acid is either converted into ethyl alcohol or lactic acid.
- The chemical equation for anaerobic respiration is:



Anaerobic respiration can be classified into several types based on the type of end product of respiration. *Some important types of anaerobic respiration are:*

- **Alcoholic fermentation:** Glucose is split into **ethanol and carbon dioxide**, producing two ATP molecules. This type is common in **yeast and some bacteria**.
- **Lactic acid fermentation:** Glucose is split into two molecules of lactic acid, producing two ATP molecules. Occurs in some bacteria and in animal muscles during strenuous exercise.
- **Methanogenesis:** Carbon dioxide is reduced to **methane by hydrogen or organic compounds**, producing one or fewer ATP molecules. This type of anaerobic respiration is performed by some archaea, such as methanogens.
- **Acetogenesis:** Carbon dioxide is reduced to **acetic acid by hydrogen or organic compounds**, producing one ATP molecule. This type of anaerobic respiration is performed by some bacteria, such as **acetogens**.

3.3 Aerobic Vs Anaerobic Respiration

	Aerobic	Anaerobic
Reactants	Glucose and oxygen	Glucose
Products	ATP, water, and CO ₂	ATP and lactic acid (animals); or ATP and ethanol (yeast)
Location	Cytoplasm (glycolysis) and mitochondria	Cytoplasm
Stages	Glycolysis (anaerobic), Krebs cycle, oxidative phosphorylation	Glycolysis, fermentation
Energy produced	Large amount (36 ATP)	Small amount (2 ATP)

BCW Facts:

- **Insects** respire through a system of air-filled tubes referred to as **tracheae**. In cockroaches, as well as numerous other insects, these tubes link to the external environment through tiny openings located along the sides of the body, which are termed **spiracles**. Upon inhalation, the spiracles open, permitting oxygen-rich air to enter the tracheae directly, thereby supplying oxygen to the tissues.
- **Coronaviruses** represent a category of interconnected **RNA viruses** that lead to illnesses in both mammals and birds. In humans and avians, they result in respiratory tract infections such as **SARS (Severe Acute Respiratory Syndrome)**, **MERS (Middle East Respiratory Syndrome)**, and **COVID-19 (Coronavirus Disease 2019)**. **Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)** is a variant of coronavirus that is the cause of the **COVID-19 pandemic**.

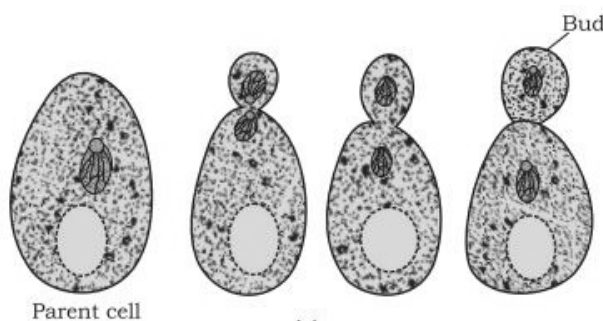
11- REPRODUCTIVE SYSTEM

1. Reproduction

- **Reproduction** is the biological process by which organisms produce offspring. The offspring grow, mature, and reproduce, ensuring continuity through a cycle of **birth, growth, reproduction, and death**.
- Reproduction may occur in two ways: (i) **Asexual reproduction** and (ii) **Sexual reproduction**.

1.1 Asexual Reproduction

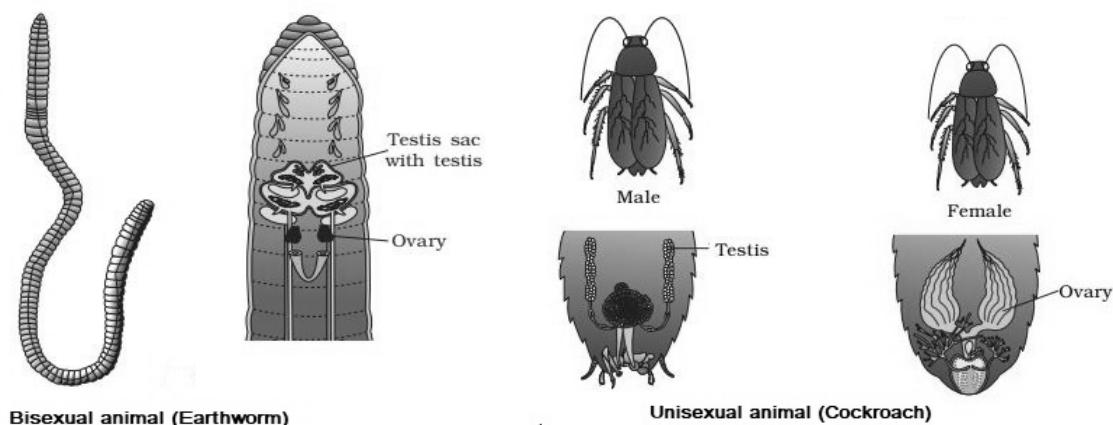
- A mode of reproduction in which a **single parent** produces offspring **without the formation of gametes**.
- It is a common process of reproduction in **single-celled organisms**, and in plants and animals with relatively simple organisation.
- The offspring produced are generally **identical to the parent**.
- Some examples are **binary fission, budding, fragmentation, and sporogenesis**.



Cell division in unicellular organism: Budding in yeast

1.2 Sexual Reproduction

- Sexual reproduction is the **production of offspring by the formation and subsequent fusion of gametes**.
- At fertilization, **the male and the female gametes unite to form a zygote**, which develops into a mature organism.
- It is the primary mode of reproduction in **most animals and higher plants**.
- Organisms may be **bisexual** or **unisexual**.
 - **Bisexual animals:** Earthworm, sponge, tapeworm, leech.
 - **Unisexual animals:** Humans, dogs, tigers, cockroaches.
- In plants, bisexual and unisexual conditions are termed **monoecious** and **dioecious**, respectively; flowers may also be **bisexual** or **unisexual**.



2. Human Reproductive System

- The **reproductive system** has four main functions: production of gametes (sperms and ova), transport and maintenance of gametes, nourishment of the developing offspring, and hormone production.

- Human reproductive events include **gametogenesis** (formation of sperms and ova), **insemination** (transfer of sperms into the female genital tract), and **fertilization** (fusion of gametes to form a zygote).
- This is followed by **blastocyst formation**, **implantation** in the uterine wall, **embryonic development (gestation)**, and **delivery of the baby (parturition)**.

2.1 The Male Reproductive System

The male reproductive system is located in the **pelvis region** and consists of a **pair of testes along with accessory ducts, accessory glands, and the external genitalia (penis)**. The testes are housed within sac-like structures known as the **scrotum**.

- From each testis, a tube called the **vas deferens**. This tube connects to the **urethra**.
- The urethra is enclosed in a muscular organ called the penis, which features an opening known as the **urethral meatus**.
- The **seminal vesicles and prostate gland produce fluids** that provide nutrition and facilitate sperm motility.
- The **process of spermatogenesis** occurs within the seminiferous tubules.
- The **Sertoli cells** are the **supporting cells that provide nutrients** to the developing sperm.
- The **Leydig cells**, which are polyhedral in shape, are located between the seminiferous tubules and secrete testosterone. It initiates the process of spermatogenesis.

A. Testes

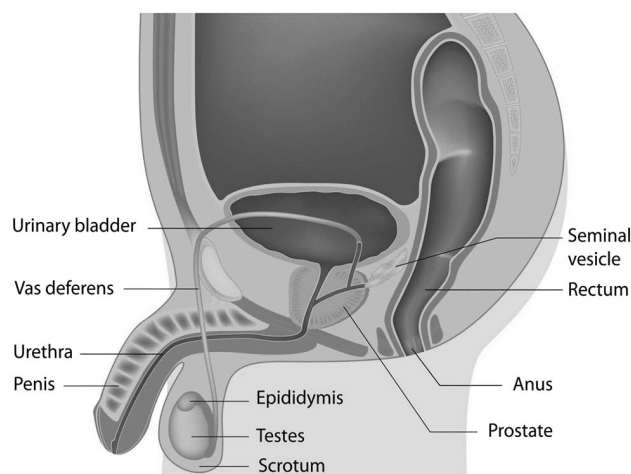
- The **testes** are located outside the abdominal cavity in the **scrotum**, which maintains a temperature **2–2.5°C below body temperature**, essential for spermatogenesis.
- Each testis contains about **250 testicular lobules**.
- Each lobule has **1–3 highly coiled seminiferous tubules**, the site of sperm production.
- Seminiferous tubules are lined by **spermatogonia (male germ cells)** and **Sertoli cells**.

B. Accessory Ducts

- Male accessory ducts include the **rete testis, vasa efferentia, epididymis, and vas deferens**.
- Seminiferous tubules within the testes open into the **vasa efferentia** through the **rete testis**.
- Vasa efferentia open into the **epididymis**, located on the posterior surface of each testis.
- The **epididymis** continues as the **vas deferens**, which loops over the urinary bladder.
- The vas deferens receives a duct from the seminal vesicle and opens into the urethra as the **ejaculatory duct**.
- These ducts **store and transport sperm** to the exterior through the **urethra**.
- The **urethra** extends from the urinary bladder through the penis and opens at the **urethral meatus**.

C. Accessory Glands

- Male accessory glands include **paired seminal vesicles, bulbourethral (Cowper's) glands, and a single prostate gland**.
- Seminal vesicles** secrete alkaline **seminal plasma** containing fructose, ascorbic acid, prostaglandins, and **vesiculase**, which enhances sperm motility.
- Bulbourethral glands** secrete fluid that lubricates the penis during copulation.
- The **prostate gland**, located below the urinary bladder and surrounding the urethra, secretes a slightly acidic fluid containing citrate, enzymes, and **prostate-specific antigens (PSA)**.



- **Semen (seminal fluid)** is a milky-white fluid consisting of **sperm and seminal plasma**.

D. Penis

- The **penis** is the male external genital organ and contains erectile tissue that facilitates insemination.
- Its enlarged terminal part, the **glans penis**, is covered by a loose fold of skin called the **foreskin**.

Organ	Function
Seminiferous tubules in testes	Produce sperms
Epididymis	Stores sperms in a viable but immobile state
Sperm duct (vas deferens)	Contractions help in the passage of sperms into urethra during ejaculation
Seminal vesicles and prostate gland	Secrete fluid, which activates and nourishes sperms
Urethra	Contractions expel semen from the penis during ejaculation (urethra also carries urine to exterior)
Penis	Contains spongy tissue and serves as the copulatory organ

2.2 The Female Reproductive System

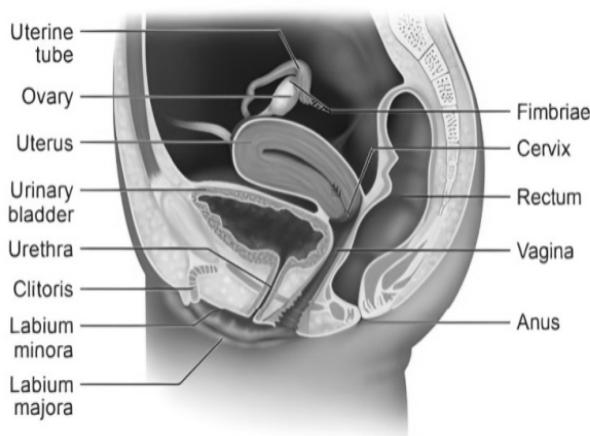
The female reproductive system consists of a **pair of ovaries along with a pair of oviducts, uterus, cervix, vagina, and the external genitalia located in the pelvic region**. A pair of the **mammary glands** is integrated structurally and functionally to support the processes of **ovulation, fertilisation, pregnancy, birth, and child care**.

A. Ovaries

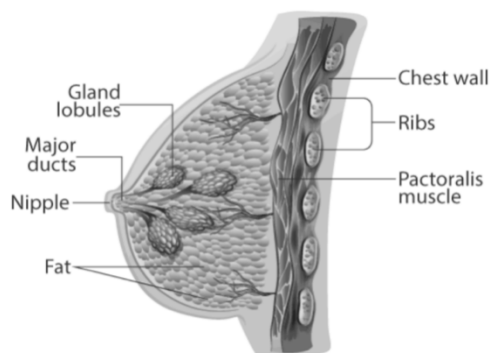
- **Ovaries** are the primary female sex organs that produce **the female gamete (ovum)** and **ovarian hormones**.
- They are located on either side of the lower abdomen and are attached to the pelvic wall and uterus by **ligaments**.
- Each ovary is covered by a thin epithelium enclosing the **ovarian stroma**, which is differentiated into an outer **cortex** and inner **medulla**.

B. Female accessory ducts

- The female accessory ducts include the **fallopian tubes (oviducts), uterus, and vagina**.
- Each fallopian tube is about **10–12 cm long** and extends from the ovary to the uterus; its terminal **isthmus** has a narrow lumen and joins the uterus.
- The **uterus** is a single, inverted pear-shaped organ supported by pelvic ligaments and opens into the vagina through the **cervix**.
- The **cervical canal** and vagina together form the **birth canal**.
- The uterine wall consists of:
 - **Perimetrium** – outer thin membranous layer
 - **Myometrium** – middle muscular layer

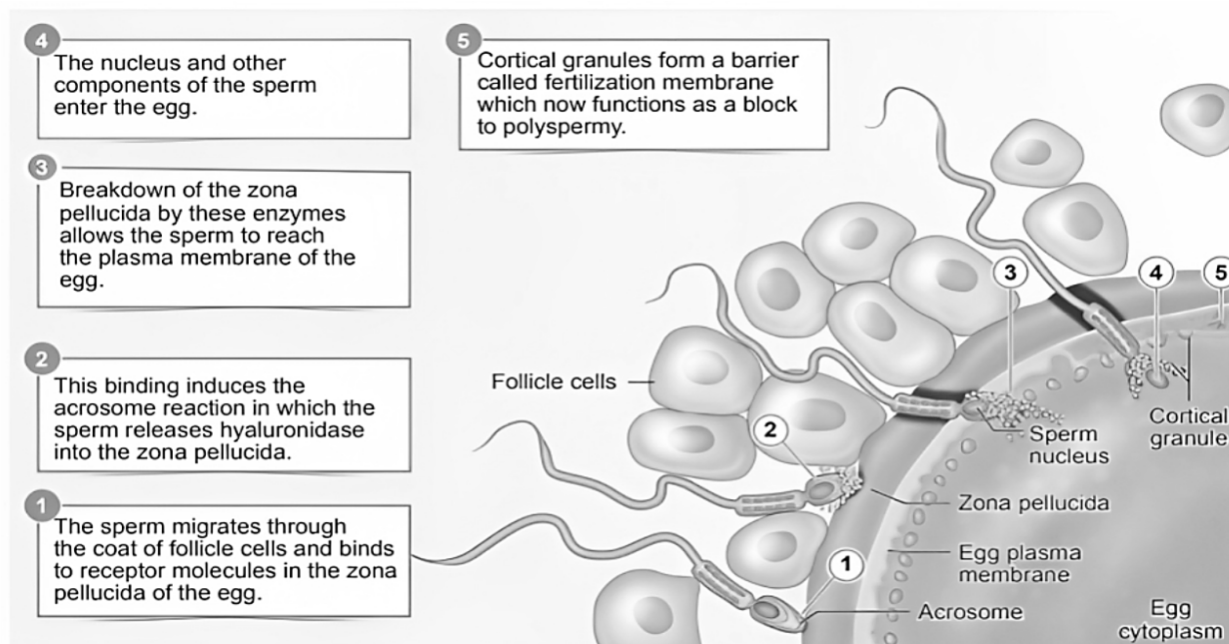


- **Endometrium** – inner glandular layer
- The **endometrium** undergoes cyclic changes during the menstrual cycle, while the **myometrium** contracts strongly during childbirth.
- C. **The mammary glands**
 - A functional mammary gland is **characteristic of all female mammals**. **Mammary glands (breasts)** are paired structures containing glandular tissue and varying amounts of fat.
 - Each breast has **15–20 mammary lobes** containing milk-secreting **alveoli**.
 - Milk is stored in the lumens of alveoli and passes into **mammary tubules**.
 - Tubules unite to form **mammary ducts**, which join into a **mammary ampulla**.
 - The ampulla connects to the **lactiferous duct**, through which milk is released.



Organ	Function
Ovary	Manufactures eggs
Oviduct (or Fallopian tube)	Site of fertilization; transfers fertilized egg/embryo to the uterus
Uterus	Inner lining receives, protects & nourishes embryo; contractions of muscular wall expel baby during birth
Cervix (neck of uterus)	Produces watery mucus that serves as a lubricant for the penis and as a medium in which sperms swim after ejection
Vagina	Receives penis during intercourse; passage for baby during birth
Clitoris (external sensual organ)	Equivalent to the male penis

3. The Reproductive Events



A. Gametogenesis

- **Gametogenesis** is the process of gamete formation in the primary sex organs—**testes** and **ovaries**.
- **Spermatogenesis** produces **sperm** in the seminiferous tubules of the testes.
- **Oogenesis** is the formation of **ova (eggs)** in the ovaries

B. Menstrual Cycle in Females

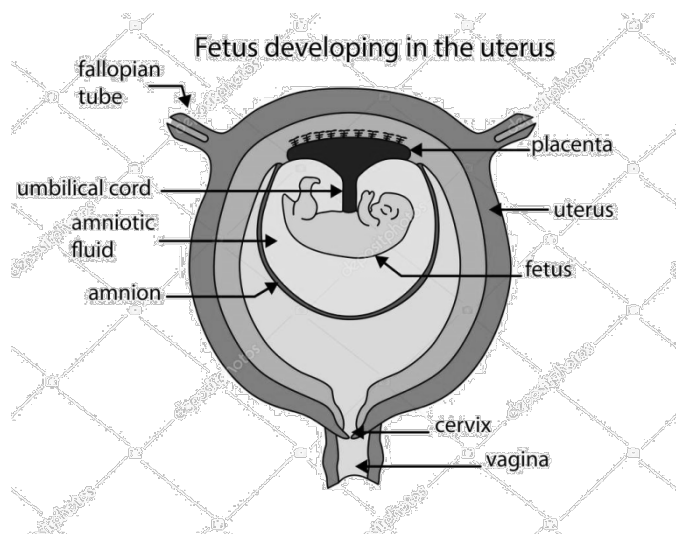
- The female reproductive phase extends from **puberty (12–13 years)** to **menopause (45–50 years)**.
- The **menstrual cycle** is a monthly sequence of events, usually lasting **28–29 days**, during which an ovum matures and is released.
- However, due to various factors, the cycle length may increase or decrease. From the 5th up to the 13th day of the onset of the menstrual cycle, growth and **maturation of the Graafian follicle take place**. The **Graafian follicle** is the **final stage in the maturation of an ovum inside the ovary**.
- **Ovulation** is followed by the **luteal phase**, during which the Graafian follicle transforms into the **corpus luteum**.
- The corpus luteum secretes **progesterone**, which maintains the **endometrium** for implantation and pregnancy.
- If fertilization does not occur, the corpus luteum degenerates, the endometrium breaks down, and **menstruation** begins.
- The menstrual cycle ceases around 50 years of age; this is termed **menopause**.

C. Fertilisation and Implantation

- **Fertilisation** is the fusion of a sperm and an ovum.
- It occurs only if the **ovum** and **sperm** are transported simultaneously to the **ampullary-isthmic junction** of the fallopian tube.
- The resulting **zygote** undergoes division, moves to the uterus, and attaches to the uterine wall.
- This attachment is called **implantation**, occurring about **one week after fertilization**.

D. Pregnancy and embryonic development

- The developing embryo (**foetus**) is connected to the uterus by the **placenta**.
- The placenta supplies **oxygen, nutrients, and antibodies** to the foetus, and also transports **carbon dioxide and wastes** from the foetal blood to the maternal blood.
- The **placenta** is connected to the embryo by the **umbilical cord**.
- The embryo is enclosed in the **amnion**, which contains **amniotic fluid** that protects it from shock and injury.
- The placenta acts as an endocrine tissue and secretes several hormones like **human chorionic gonadotropin (hCG)**, **human placental lactogen (hPL)**, **estrogens**, and **progestogens**.
- In later pregnancy, the ovary secretes **relaxin**.
- **hCG, hPL, and relaxin** are produced only during pregnancy.



E. Parturition and lactation

- **Parturition** is the process of childbirth.
- Signals from the fully developed foetus and placenta initiate the **foetal ejection reflex**, leading to **oxytocin** release.
- **Oxytocin** (from the posterior pituitary) stimulates uterine contractions during childbirth.
- **Lactation** is the secretion of milk by the mammary glands.
- The first milk, called **colostrum**, is rich in **nutrients, fats, and protein**. Colostrum also contains **antibodies (Immunoglobulin A-IgA)** that provide passive immunity to the newborn infant.
- **Prolactin** (anterior pituitary) hormone stimulates milk production, while **oxytocin** promotes milk ejection.

4. New Medical Techniques In Reproduction

There are many new medical techniques in the field of reproduction to help infertile males and females produce babies. Some of these are given here.

A. In-vitro fertilization (IVF) or Test tube baby

- Used when the **fallopian tubes are blocked, damaged, or absent**.
- Ova and sperm are fertilized **outside the body** under laboratory conditions. In this technique, **ripe ova** are collected from the woman's ovaries and cultured with the male partner's **sperm** under optimal conditions, where fertilization occurs to form an **early embryo**.
- The resulting **early embryo** is transferred to the uterus for implantation and development.

B. Gamete Intra-Fallopian Transfer (GIFT)

- An **ovum** collected from a donor is transferred into the fallopian tube of a female who cannot produce ova.
- Eggs and sperm are placed in the fallopian tube, where fertilization occurs naturally.
- The zygote moves to the uterus for implantation.

C. Intra cytoplasmic Sperm Injection (ICSI)

- Another specialized technique in which a **single sperm is directly injected into an ovum** to form an embryo.

D. Zygote Intra-Fallopian Transfer (ZIFT) and IVF

- Fertilization occurs **in the laboratory**.
- The resulting **zygote** is transferred into the fallopian tube using laparoscopy and later implants in the uterus.

E. Intra-Uterine Insemination (IUI)

- Used in cases of **low sperm count** or inability to inseminate. It is used to treat infertility caused by the inability to inseminate or a very low sperm count. In this **Artificial insemination (AI)** technique, semen from the husband or a healthy donor is artificially introduced into the **vagina or uterus** of the female.

F. Intra-Uterine Transfer (IUT)

- An **embryo with more than 8 blastomeres** is transferred into the uterus for further development.

G. Surrogacy

- **Surrogacy** is a method of **assisted reproduction or agreement whereby a woman agrees to carry a pregnancy for another person, who will become the newborn child's parent after birth**.
- Through *in vitro* fertilization (IVF), embryos are created in a lab and transferred into the surrogate mother's uterus.

Cloning

- **Cloning** refers to the creation of an identical copy of a cell, any other living part, or an entire organism. It is a key technique in **biotechnology** that allows the replication of biological material.
- The first successful cloning of an animal was achieved by **Ian Wilmut** and his team at the Roslin Institute located in Edinburgh, Scotland. They successfully cloned a sheep named Dolly. Dolly was born on July 5, 1996,

and was recognized as the **first mammal to be cloned**. Dolly passed away on February 14, 2003, as a result of a specific lung disease.

BCW Facts:

- **Embryogenesis** is the term used to describe the **development of an embryo from a zygote**.
- Common **sexually transmitted diseases (STDs)** include **chlamydia, genital warts, gonorrhoea, genital herpes, syphilis, trichomoniasis, hepatitis B, and HIV/AIDS**.
- The male sterilization procedure is referred to as '**vasectomy**,' while the female procedure is called '**tubectomy**.' In a vasectomy, a small section of the vas deferens is either removed or tied off through a small incision in the scrotum, whereas in a tubectomy, a small portion of the fallopian tube is removed or tied off through a small incision in the abdomen or vagina.
- **Lactational amenorrhea (LAM)** is a natural and temporary phase of infertility experienced by breastfeeding women, marked by the absence of menstruation.

12 - MUSCULAR AND SKELETAL SYSTEM

1. Muscular System

- The **muscular system** enables movement of the limbs, jaws, tongue, and other body parts through **muscle contraction**.
- Muscular tissue** consists of elongated cells called **muscle fibres**, which contain **contractile proteins** responsible for contraction and relaxation.
- The human body has **over 600 muscles**, performing functions ranging from **locomotion** to fine movements like threading a needle.

2. Muscle And Its Types

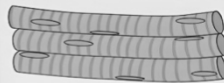
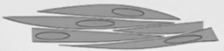
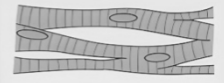
- Muscle** is a specialized tissue derived from the **mesoderm** and constitutes about **40–50% of an adult's body weight**.
- Its key properties are **contractility, excitability, extensibility, and elasticity**.
- Based on location, muscles are of **three types**:
 - Skeletal
 - Visceral (Smooth)
 - Cardiac

2.1 Skeletal Muscles

- Attached to the **skeleton** and appear **striated** under the microscope.
- Also called **voluntary muscles**, as they are under the voluntary control of the nervous system.
- Responsible for **locomotion** and **maintaining body posture**.

2.2 Visceral (Smooth) Muscles

- Found in the walls of **hollow visceral organs** such as the **alimentary canal** and **reproductive tract**.
- They are **non-striated (smooth)** and **involuntary**.
- Help in the movement of **food through the digestive tract** and **gametes through the reproductive tract**.

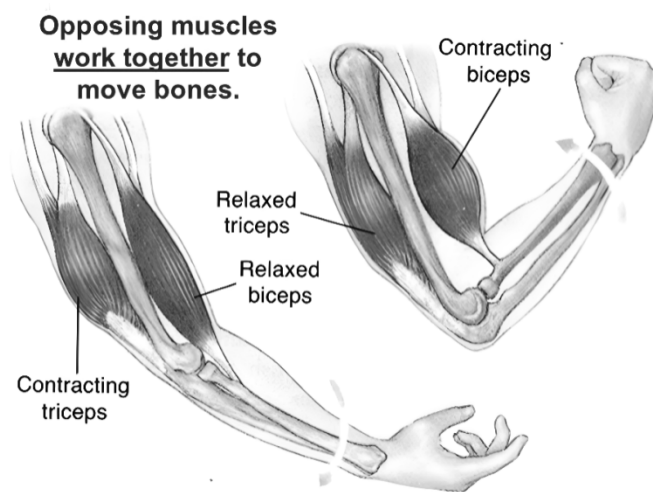
	Main features	Histology
Skeletal muscle	- Fibers: striated, tubular and multi nucleated - Voluntary - Usually attached to skeleton	
Smooth muscle	- Fibers: non-striated, spindle-shaped, and uninucleated - Involuntary - Usually covering wall of internal organs	
Cardiac muscle	- Fibers: striated, branched and uninucleated - Involuntary - Only covering walls of the heart	

2.3 Cardiac Muscles

- Found **only in the heart**.
- Composed of **branched, interconnected, striated muscle fibres**.
- Function **involuntarily** and are responsible for the **rhythmic contraction of the heart**.

3. Structure And Mechanism Of Muscles Movement

- Each **skeletal muscle** is composed of **muscle bundles (fascicles)** enclosed by a collagenous connective tissue layer called **fascia**.
- Each **fascicle** contains numerous **muscle fibres**.
- Muscle contraction involves **actin (thin)** and **myosin (thick)** filaments arranged in repeating units called **sarcomeres**, the functional units of muscle.
- During **contraction**, muscles become **shorter, thicker, and stiffer**; during **relaxation**, they become **longer and thinner**.
- Muscles work in **antagonistic pairs (flexors and extensors)**. When one contracts, the other relaxes. Since, muscles can only pull, opposite movements require alternate contraction and relaxation.



- **BCW Facts:** Repeated activation of muscles can lead to the **accumulation of lactic acid** due to the anaerobic breakdown of glycogen in them, causing fatigue.

4. Red Fibres And White Fibres

- **Myoglobin** is a red, oxygen-storing pigment present in muscles.
- Muscles with **high myoglobin** content appear **red (red fibres)**, contain **numerous mitochondria**, and produce ATP **aerobically**; hence, they are called **aerobic muscles**.
- Muscles with **low myoglobin** content appear **white (white fibres)**, contain **fewer mitochondria** but abundant **sarcoplasmic reticulum**, and produce energy mainly through **anaerobic respiration**.

5. Types Of Muscle Contractions

There are three types of muscle contractions:

1.1. Isometric contraction:

- Tension is generated during the process of contraction, but there is no visible change in the length of the muscle.
- Example: pushing a wall, planks, etc.

1.2. Isotonic contraction:

- When the muscle's shortening and lengthening processes are visible while lifting a constant load.
- Example: bicep curl, push-ups.

1.3. Iso-kinetic contraction:

- It occurs when the velocity of muscle contraction stays the same while the muscle length varies.
- Example: Performing a knee extension exercise on an isokinetic dynamometer.

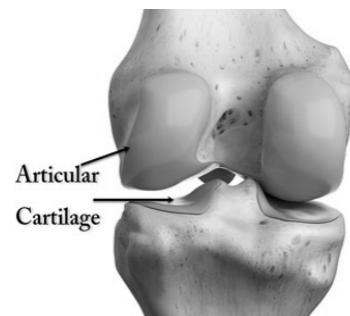
6. Bone, Cartilage, Ligament, Tendon, and Interstitial Fluid

Bone

- **Bone** is the basic unit of the human skeletal system, providing **support, body framework, protection of vital organs, weight bearing, movement, and blood cell formation (hematopoiesis)**.
- It is a **strong, rigid connective tissue** with bone cells embedded in a hard matrix rich in **calcium and phosphorus salts**.
- Bones are moved by the **alternate contraction and relaxation of antagonistic muscles**.
- The study of bones is called **Osteology**.

Cartilage

- **Cartilage** is a **flexible connective tissue** found where support with flexibility is required.
- Cartilage contains widely spaced cells, and its **solid matrix is composed of proteins and sugars**.
- It is present in the **joints, knees, ankles, ends of ribs, between vertebrae, ears, nose, trachea, and larynx**.
- Cartilage is made up of specialized cells called **chondrocytes**.



Ligament

- A type of connective tissue that **connects bones to other bones and assists in their stability and movement**.

Tendon

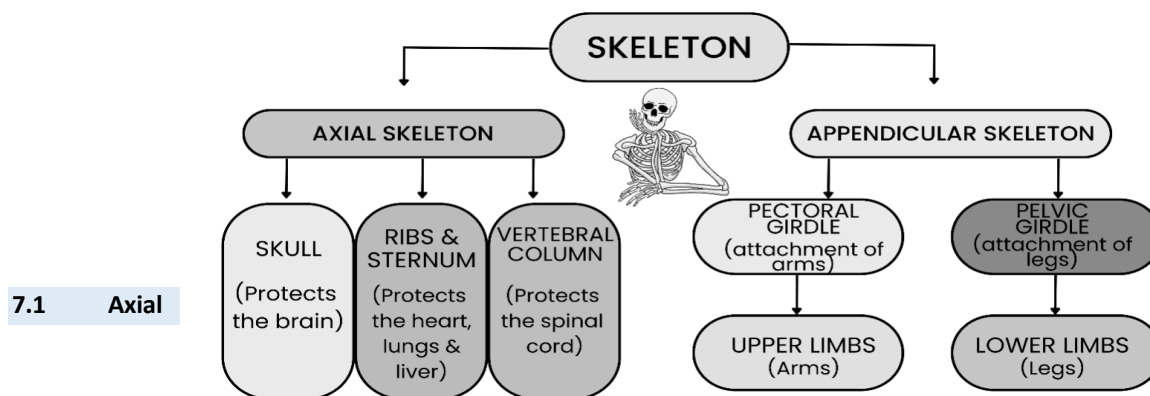
- A type of fibrous connective tissue that **connects bone to muscles and provides structural strength to them**.

Interstitial fluid

- A **fluid present in the space around cells** that facilitates the significant exchange of substances like oxygen, salts and other nutrients between blood and tissues. It is also called **tissue fluid**.

7. Skeletal System

- **The skeletal system consists of a framework of bones and a few cartilages.** It plays a significant role in body movement. Bone and cartilage are specialised connective tissues.
- Bones have a very hard matrix due to calcium salts in it, while cartilages have a slightly pliable matrix due to chondroitin salts.
- In human beings, this system is made up of **206 bones** and a few cartilages. It is grouped into **two principal divisions – the axial and the appendicular skeleton**.
- The skeleton comprises the skull, backbone, ribs and the breast bone, shoulder and hip bones, and the bones of limbs (hands and legs).

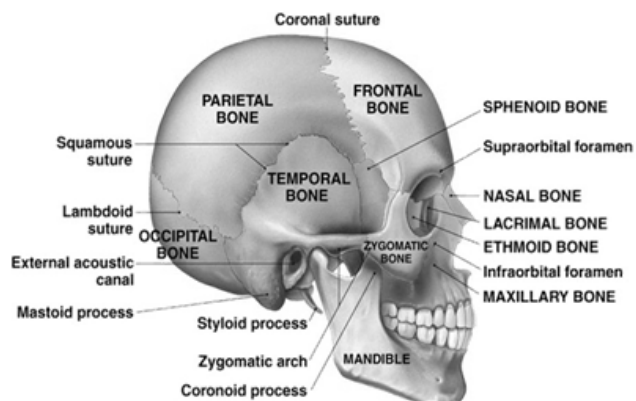


Skeleton

The axial skeleton comprises **80 bones** aligned along the central axis of the body. It includes the **skull, vertebral column, sternum, and ribs.**

A. Skull

- The **human skull** consists of **22 bones: 8 cranial and 14 facial bones.**
- The **8 cranial bones** form the **cranium**, which protects the brain.
- The **14 facial bones** form the **facial skeleton.**
- The **hyoid** is a **U-shaped bone** at the base of the buccal cavity that aids in **speech and swallowing.**
- The **mandible** is the **only movable bone** of the skull.
- Each middle ear contains **three tiny bones – Malleus, Incus and Stapes, collectively called Ear Ossicles.**

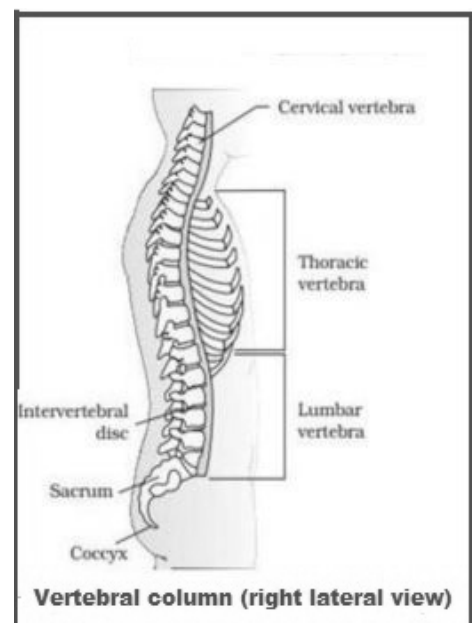


B. Vertebral Column

- The **vertebral column (backbone)** consists of **26 vertebrae** and extends from the **base of the skull**, forming the main framework of the trunk.
- It is divided into **5 regions: Cervical (7), Thoracic (12), Lumbar (5), Sacral (1 fused), and Coccygeal (1 fused).**
- Cartilaginous discs** between adjacent vertebrae provide **flexibility** and allow bending.
- The **7 cervical vertebrae** are common to **almost all mammals**, including humans.
- The vertebral column **protects the spinal cord**, **supports the head**, and provides **attachment for the ribs and back muscles.**

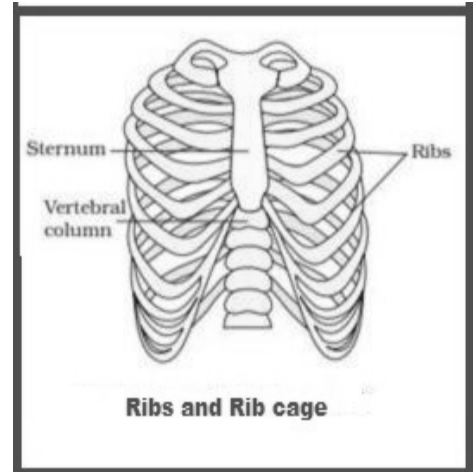
C. Sternum

- The **sternum (breastbone)** is a **flat bone** located on the **ventral midline of the thorax.**



D. Ribs

- Humans have **12 pairs (24) of ribs**, each attached **dorsally to the vertebral column**.
- **True ribs (1st–7th pairs)**: Directly attached to the **sternum** by **hyaline cartilage**.
- **False ribs (8th–10th pairs)**: Indirectly attached to the sternum through the **7th rib**.
- **Floating ribs (11th–12th pairs)**: **Not attached ventrally** to the sternum.
- The **thoracic vertebrae, ribs, and sternum** together form the **rib cage**.



7.2. Appendicular Skeleton

- The **appendicular skeleton** comprises the **bones of the limbs** and their **pectoral and pelvic girdles**.
- **Each limb contains 30 bones**.
- The **pectoral girdle** connects the **upper limbs**, while the **pelvic girdle** connects the **lower limbs** to the **axial skeleton**.
- Each girdle consists of **two halves**.

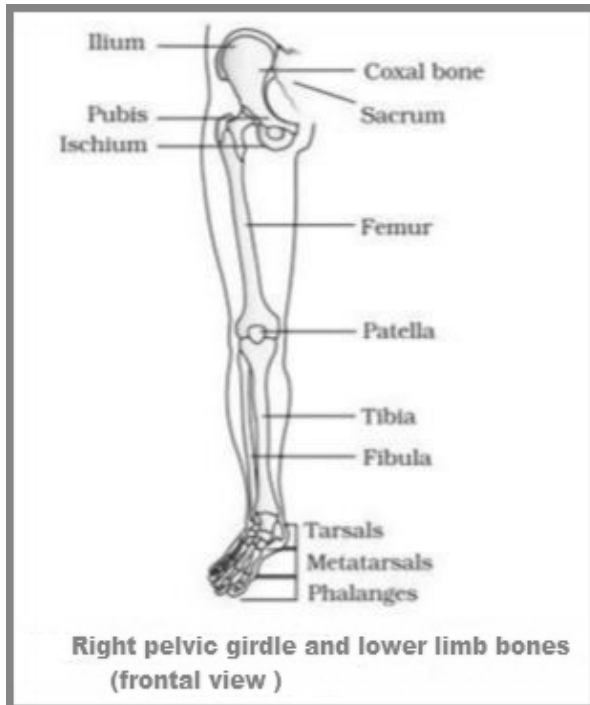
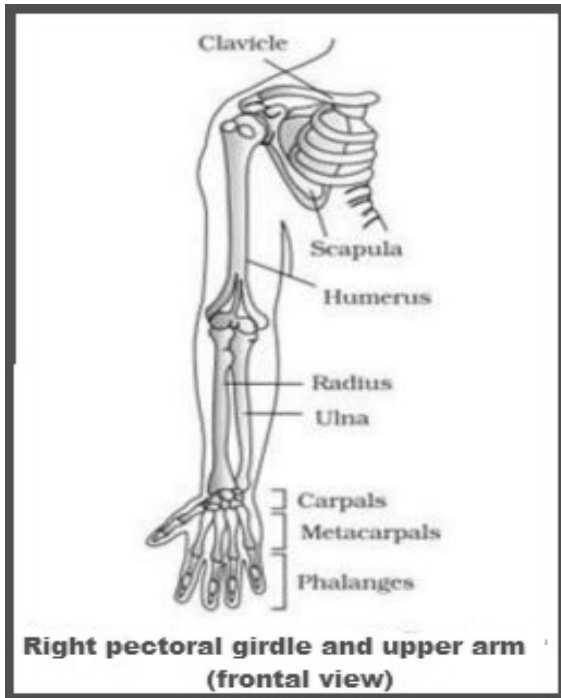
A. Arm bone

- The arm bone is the upper limb made up of the **humerus (upper arm)**, **radius and ulna (Forearm)**, **carpals (wrist bones - 8)**, **metacarpals (palm bones - 5)**, and **phalanges (finger bones -14)**.
- All these **bones are joined by hinge joints, which allow the limb to move only in one direction**.
- The wrist is flexible due to the movable joint between the carpals and forearm bones.
- The palm bones (metacarpals) form **movable joints** with the fingers.
- Finger bones (Phalanges) form movable joints among themselves for flexibility.

B. Leg bone

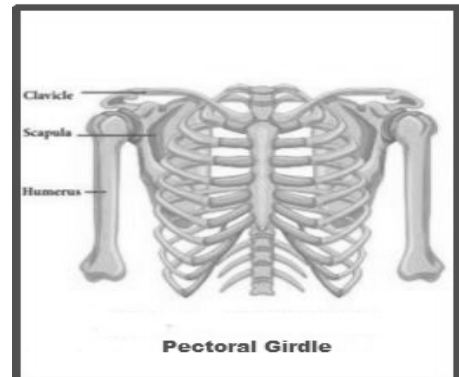
- The **leg bone is the lower limb** made up of the **femur** (thigh bone; the longest and heaviest bone), **tibia and fibula** (leg), **tarsals** (ankle bones - 7), **metatarsals** (foot bones - 5), and **phalanges** (toe bones - 14).
- **All these bones are joined by hinge joints**, which allow the limb to move only in one direction.

- The knee is protected by a cap-like structure called the **patella** or the knee cap.



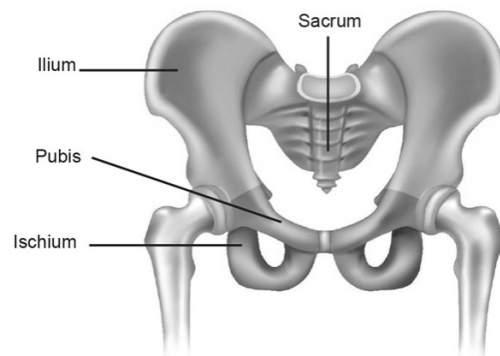
C. Pectoral girdle

- Each **pectoral girdle** consists of a **clavicle** (collarbone) and a **scapula** (shoulder blade).
- The **scapula** is a **flat, triangular bone** located on the **dorsal thorax** between the **2nd and 7th ribs**.
- It bears a prominent ridge called the **spine**, which extends into a flat, expanded process called the **acromion**.
- The **clavicle** articulates with the **acromion**.
- The **glenoid cavity** of the scapula articulates with the **head of the humerus** to form the **shoulder joint**.



D. Pelvic girdle

- The **pelvic girdle** consists of **two coxal (hip) bones**.
- Each **coxal bone** is formed by the fusion of the **ilium**, **ischium**, and **pubis**.
- The **acetabulum** is the socket that articulates with the **head of the femur (thigh bone)** to form the **hip joint**.
- The two coxal bones meet anteriorly at the **pubic symphysis**, a **fibrocartilaginous joint**.



7.3. Joints

- Joints** are points where **two bones or a bone and cartilage meet**, enabling body movement.
- Muscles generate force**, while **joints facilitate movement**.
- Structurally, joints are of **three types: Fibrous, Cartilaginous, and Synovial**.

Feature	Structure	Function
Ligament	A band of fibrous tissue	To connect bone to bone.
Synovial fluid	A slippery fluid with consistency of egg whites that is contained within the joint cavity.	To reduce friction between the articular cartilage in the joint.
Articular cartilage	Glassy- smooth cartilage that is spongy and covers the ends of the bones in the joint.	To absorb shock and to prevent friction between the ends of the bones in the joint.
Joint capsule	A tough fibrous tissue that has two layers, with the fibrous capsule lying outside the synovial membrane.	The fibrous capsule helps to strengthen the joint, while the synovial membrane lines the joint and secretes synovial fluid.

a. Fibrous Joints

- Fibrous joints permit **no movement** whatsoever. An example of this type of joint can be seen in the flat bones of the skull, which are fused end-to-end by **dense fibrous connective** tissues in the form of sutures, thereby forming the cranium.

b. Cartilaginous Joints

- Bones are connected by **cartilage**.
- The joint found between adjacent vertebrae in the vertebral column exemplifies this type and allows for **limited movement**.

c. Synovial joints

- Characterized by a **fluid-filled synovial cavity** between articulating bones.
- Allow **free movement** and are essential for **locomotion**.
- Examples include the **ball and socket joint** (located between the humerus and the pectoral girdle), **hinge joint** (as seen in the knee joint), **pivot joint** (found between the atlas and axis), **gliding joint** (between the carpals), and **saddle joint** (between the carpal and metacarpal of the thumb).

Example of Movable joints

Joint	Examples
Ball and socket joint	Shoulder, Hip
Hinge joint	Elbow, knee, Ankle
Pivot joint	Spine (Atlas/Axis joint on top)
Gliding joint	Spine (Between the bony processes of the vertebrae)
Saddle joint	Thumb, shoulder, and inner ear
Condylloid Joint	Wrist

8. Muscular and Skeletal Disorders

- Myasthenia gravis:** An autoimmune disorder affecting the neuromuscular junction, causing **muscle weakness, fatigue, and paralysis of skeletal muscles**.
- Muscular dystrophy:** A hereditary genetic disorder characterized by **progressive degeneration and weakening of skeletal muscles**, leading to immobility.

- **Arthritis:** Inflammation of the joints, causing **pain and stiffness**. **Rheumatoid arthritis** is a chronic, debilitating form associated with persistent joint pain.
- **Osteoporosis:** An age-related disorder marked by **decreased bone mass** and increased fracture risk, often linked to reduced **estrogen** levels.
- **Gout:** Inflammation of joints caused by the **deposition of uric acid crystals**.
- **Tetany:** Characterized by **rapid muscle spasms** due to **low calcium levels** in body fluids.

BCW Facts:

- The **stapes** is the **smallest bone of the body** constitutes the innermost bone of the middle ear.
- **The tibia** is a weight-bearing bone of the leg and the second largest bone of the human body.
- An **earthworm** does not have bones. It has **muscles that help to extend** and shorten the body. Earthworms move by alternate extension and contraction of the body using muscles.
- At birth, a **baby possesses approximately 300 bones**. These bones eventually fuse (grow together) to create the 206 bones found in adults.
- The **stapedius** is the **smallest skeletal muscle** in the human body, situated in the middle ear. Its primary role is to stabilize the stapes bone (the smallest bone in the body) and regulate the amplitude of sound waves, thereby protecting the inner ear from excessive noise.

13 - EVOLUTION

1. What Is Evolution?

- **Evolution** is the process of **heritable changes in the characteristics of a population over successive generations**.
- **Origin of life** refers to the emergence of the **first primitive life forms from non-living matter**.
- **Evolution of life** is the gradual development of **complex organisms from simpler forms** over time.
- Understanding the evolution of **flora and fauna** requires knowledge of the origin of life, i.e., the evolution of earth, of stars, and indeed, of the universe itself.



2. Origin Of Life

- The origin of life is considered a unique event in the history of the universe.
- The vast universe is very old – **almost 13.8 billion years old**. The **Big Bang theory** attempts to explain to us the origin of the universe. Earth is believed to have formed about **4.5 billion years ago**. There was **no atmosphere on early earth**.
- The early Earth lacked an atmosphere and was surrounded by **water vapour, methane, carbon dioxide, and ammonia** released from the molten surface.
- **Life appeared 500 million years** after the formation of earth, i.e., almost four billion years ago.

Evolution of Life Forms – A Theory

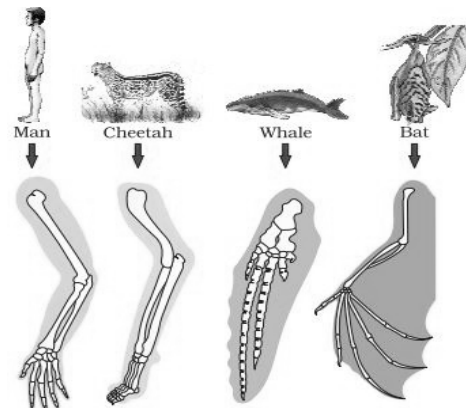
- The “**Theory of special creation**” states that life was created by a **supernatural power**.
- According to the **Theory of Abiogenesis (proposed by Thomas Henry Huxley)**, living organisms originated from non-living materials and occurred through stepwise chemical and molecular evolution over millions of years.
- According to the **Theory of biogenesis**, life arose **from pre-existing life**. The term biogenesis also refers to the biochemical process involved in the production of living organisms.
- According to the **Theory of chemical evolution**, primitive organisms in the primordial environment of earth evolved **spontaneously from inorganic substances and physical forces** such as lightning, UV radiation, volcanic activities, etc.
- **Alexander Oparin (1924)** suggested that the organic compounds could have undergone a series of reactions, leading to more complex molecules.

3. Evidences For Evolution

3.1. Evidences from Morphology

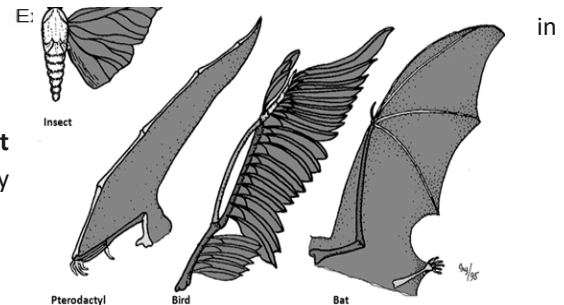
A. Homologous Organs

- **Homologous organs** have the **same origin and basic structure** but may look very different and perform different functions.
- They provide evidence of **common ancestry** and **divergent evolution**.
- **Example:** The forelimbs of **humans, whales, bats, and cheetahs** have the same basic bone pattern (**humerus, radius, ulna, carpals, metacarpals, and phalanges**) but perform different functions.
- **Plant example:** **Thorns of Bougainvillea** and **tendrils of Cucurbita** are homologous structures.



B. Analogous Organs

- **Analogous organs** perform the **same function** but differ in **origin and structure**.
- They are evidence of **convergent evolution**.
- **Example:** The **wings of insects and birds** both enable **flight** but differ in their structural organization and evolutionary origin.

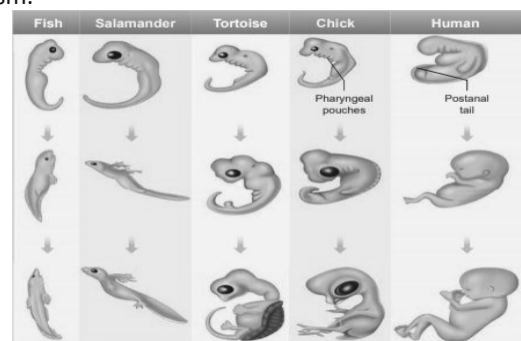


C. Vestigial organs

- **Vestigial organs** are **reduced or non-functional structures** that are no longer essential for survival.
- They were **functional in ancestral organisms** and have been inherited through evolution.
- Vestigial organs provide **evidence for evolution** and **common ancestry**.
- **Examples in humans:** **Appendix, wisdom teeth, coccyx (tailbone), body hair, and tonsils.**

3.2. Evidences from Embryology

- **Embryology** is the study of the development of an organism.
- The embryos of all vertebrates are similar in shape and structure in their early stages. We see similar stages of early development (**morula, blastula, or gastrula**) in all the animals. All vertebrates begin life from a single cell, the zygote.



3.3. Evidences from Palaeontology

- Palaeontology is the **study of fossils**. The study of fossils in different sedimentary layers indicates the geological period in which they existed.
- The study showed that **life-forms varied over time**, and certain life forms are restricted to certain geological time-spans.
- Some fossils resemble present-day organisms, while others represent **extinct species**, such as **dinosaurs**.



4. Biological Evolution Theories

A. Lamarck's Theory (Lamarckism)

- Jean-Baptiste Lamarck proposed the **first theory of evolution** in **1809** in his famous book *Philosophie Zoologique*.
- **Theory of Use and Disuse:** Frequently used organs become **well developed**, while unused organs **degenerate**.
Examples: Long neck of giraffe (use) and **limbless snakes** (disuse).
- **Inheritance of Acquired Characters:** Traits acquired during an organism's lifetime are **passed on to offspring**.
- **Objection to Lamarckism:**
 - August Weismann disproved Lamarck's *theory of acquired characters* by **cutting the tails of mice for 20 generations**; all offspring were born with tails.
 - He proposed the **Germ Plasm Theory**, stating that **only changes in germ cells are inherited**, not changes in somatic (body) cells.

B. Neo-Lamarckism

- Scientists such as **Cope, Osborn, Packard, and Spencer** supported and modified Lamarck's ideas.
- They believed that **adaptations to environmental changes** led to the development of new structures.

C. Darwin's Theory of Natural Selection

- Charles Darwin explained evolution in his book *The Origin of Species by Natural Selection* (**1859**) and is known as the **Father of Evolution**.
- **Key contributions:**
 - All organisms share **common ancestry**.
 - Evolution occurs through **natural selection**.
- Organisms produce **more offspring than can survive**.
- Due to **limited resources**, individuals compete for survival (**struggle for existence**).
- Individuals with **advantageous heritable variations** survive and reproduce (**survival of the fittest**), while less adapted individuals are eliminated.

Objections to Darwinism:

- Failed to explain the **origin of variations**.
- Explained the **survival** of the fittest, but not the arrival of the fittest.
- Emphasized **small fluctuating (mostly non-heritable) variations**.
- Could not explain **vestigial organs**.

D. Mutation Theory

- Hugo de Vries proposed the **Mutation Theory (1901)**.
- **Mutations** are **sudden, random, heritable changes**.
- Based on experiments on **Evening Primrose plant (*Oenothera lamarckiana*)**.
- Proposed that **new species arise through sudden large mutations**, unlike Darwin's gradual evolution.

E. Modern Synthetic Theory (Neo-Darwinism)

- Developed by scientists including **Sewall Wright, R.A. Fisher, Ernst Mayr, Julian Huxley, Theodosius Dobzhansky, and G.G. Simpson**.
- Evolution is driven by **five major factors**:
 - **Gene mutations:** Changes in gene structure producing heritable variations. It alters the phenotype of an organism and produces variations in its offspring.
 - **Chromosomal mutations:** Changes due to **deletion, duplication, inversion, addition, or translocation**. This, too, alters an organism's phenotype and produces variations in its offspring.
 - **Genetic recombination: Crossing over during meiosis** creates new gene combinations. This brings about genetic variations in the individuals of the same species and leads to heritable variations.
 - **Natural selection:** Favours beneficial genetic variations.

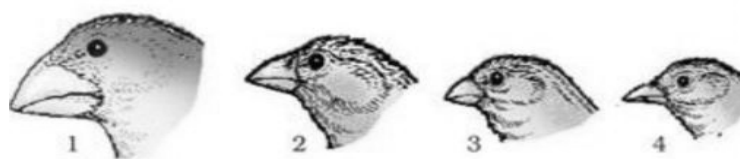
- **Reproductive isolation:** Does not produce any genetic variations, but once such variations occur, it favours some genetic changes while rejecting others.

F. Evolution by Anthropogenic Sources

- Evolution by anthropogenic sources is the process by which human activities affect the genetic variation and adaptation of living organisms. Some examples of anthropogenic sources are:
- **Industrial pollution (Industrial melanism):** Smoke and soot darkened tree bark and lichens, favouring the survival of **dark-colored moths** over light-colored ones due to better camouflage.
- **Agricultural practices:** Excessive use of **pesticides and herbicides** has led to the evolution of **resistant insects and weeds**.
- **Antibiotic use:** Overuse and misuse of antibiotics have resulted in the emergence of **multidrug-resistant bacteria**, reducing the effectiveness of antibiotics.
- **DDT-resistant mosquitoes:** Continuous use of **DDT** selects mosquitoes carrying **resistant mutations**, allowing them to survive and reproduce through **natural selection**.

G. Adaptive Radiation

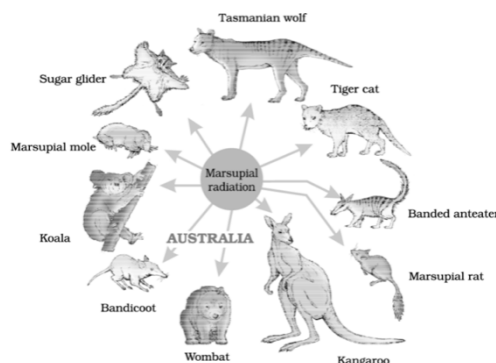
- **Adaptive radiation** is the evolution of **multiple species from a common ancestor** to occupy different habitats or ecological niches.
- **Darwin's finches** (Small black birds) of the **Galápagos Islands** are the classic example, evolving different beak types for different food sources.



Variety of beaks of finches that Darwin found in Galapagos Island

- Another example is **Australian marsupials**. Several marsupials, each distinct from one another, have evolved from a common ancestral lineage, all within the **Australian island continent**.

NOTE: A marsupial is a mammal belonging to an order characterized by members that are born in an underdeveloped state and are usually carried and nursed in a pouch located on the mother's abdomen. Marsupials are primarily located in Australia, New Guinea, and also in the Americas.

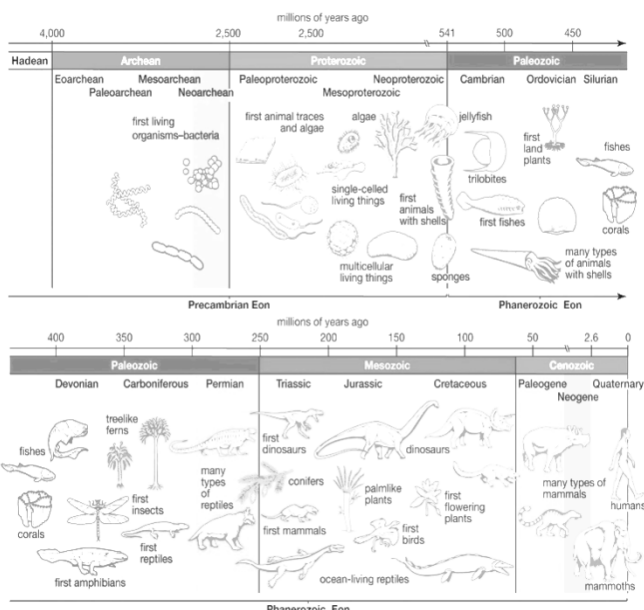


5. **A Brief Account Of Evolution**

- Approximately 2000 million years ago (mya), the first cellular life forms emerged on Earth.
- Gradually, single-celled organisms evolved into **multi-cellular life forms**. By around 500 mya, invertebrates had developed. Jawless fish likely appeared approximately 350 mya.
- Seaweeds and a few plant species are believed to have existed around 320 mya.
- Fish with robust and powerful fins were capable of moving onto land and returning to the water. This was around 350 mya.
- In the next 200 million years or so, reptiles of different shapes and sizes dominated the earth. Giant ferns (pteridophytes) were present, but they all fell to form coal deposits slowly. Some of these land reptiles went back into water to evolve into fish-like reptiles, probably 200 mya (e.g., Ichthyosaurs). The biggest of them,

i.e., Tyrannosaurus rex, was about 12 feet in height & 40 feet in length and had huge, fearsome, dagger-like teeth.

- About 65 mya, the **dinosaurs suddenly disappeared from the earth**.
- Small-sized reptiles of that era still exist today.



believed to have been able to talk), lived about 2 mya. Their brain capacity was between 650cc and 800cc, and they were probably vegetarian.

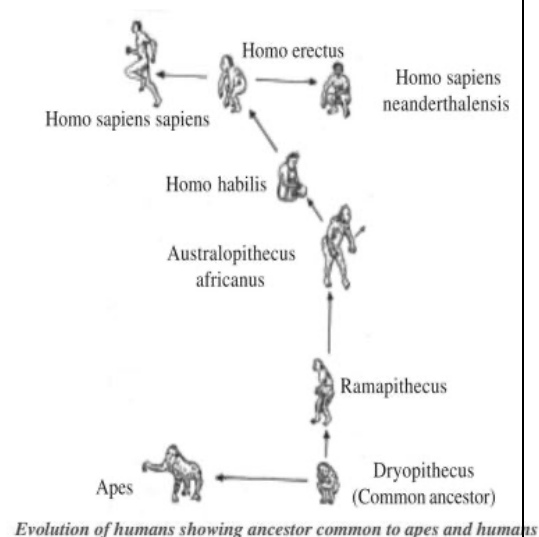
- Fossils discovered in Java in 1891 revealed the next stage, i.e., **Homo erectus**, about 1.5 mya. Homo erectus had a large brain of around 900cc and was probably non-vegetarian.
- The Neanderthal man, with a brain size of 1400 cc, lived in the Near East and Central Asia between 100,000 and 40,000 years ago. They used hides to protect their body and buried their dead.
- **Homo sapiens** arose in Africa, moved across continents, and developed into distinct races. During the ice age between 75,000 and 10,000 years ago, modern **Homo sapiens** arose.
- Prehistoric cave art developed about 18,000 years ago. Agriculture came around 10,000 years ago, and human settlements started.

BCW Facts:

- **Phenylketonuria**, a genetic disorder, is caused by a mutation in the phenylalanine hydroxylase gene, leading to **elevated levels of phenylalanine in the bloodstream**.
- **Archaeopteryx** is a connecting link between reptiles and birds (Aves). It is regarded as the earliest known bird, originating from the **Jurassic era**.
- The **extinct mammoth** is more closely related to today's Asian elephant than to the African elephant.
- Father of modern biology in India- **Dr. Pushpamitra Bhargava**

6. Origin And Evolution Of Man

- About 15 million years ago (mya), primates called Dryopithecus and Ramapithecus lived. They were hairy and walked like gorillas and chimpanzees. Ramapithecus was more man-like, while Dryopithecus was more ape-like.
- Two mya, Australopithecus probably lived in the East African grasslands. Evidence shows they hunted with stone weapons but essentially ate fruit.
- The first human-like being was called **Homo habilis**. They could walk upright. (They are



Fathers Of Various Branches Of Biology

Branches Of Biology	Father	Branches Of Biology	Father
Botany	Theophrastus	Indian Paleobotany	Birbal Sahni

Embryology	Karl Ernst Von Baer	Mutation	Hugo de Vries
Genetics	Gregor Johann Mendel	Pathology	Rudolph Virchow
Immunology	Edward Jenner	Taxonomy	Carolus Linnaeus
Indian Ecology	Ramdeo Mishra	Zoology	Aristotle